Mathematical Stories

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Lecture Notes

July 15, 2026

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Introduction

Purpose and Philosophy

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognises the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, formulate precise questions, distinguish an object from its notation, notice structure, and build simple but meaningful chains of reasoning. A solved problem is valuable not only because it produces an answer, but also because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorise, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as an answer to a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some habits of thought will deliberately return in different places. The same question of what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. A mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognise patterns of reasoning rather than only patterns of exercises.

The number of answers produced is not, by itself, a measure of progress. What matters is the quality of the work. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each important step was valid? Did we interpret or verify the result? These questions are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics as the art of asking good questions and giving well-reasoned answers.

How the Course Works

This book is part of a weekly learning process rather than a collection of material to be read only before an examination. Each of the twelve thematic chapters is designed to support one week of study. A chapter provides the mathematical structure for the lecture, a complete narrative to which the student can return, and a starting point for individual problem solving and classroom discussion.

In a fifteen-week semester, the twelve thematic weeks establish the main rhythm of the course. The remaining time can be used to introduce the method of work, connect topics, revise important ideas, and present more substantial solutions.

The Weekly Learning Cycle

A student should not leave a topic behind when the lecture ends. A typical week consists of the following connected stages:

1. **Attend the lecture.** The lecture introduces the topic, motivates the central questions, and demonstrates how the relevant mathematical objects and methods should be read.

2. **Return to the chapter.** Read the corresponding chapter carefully after the lecture. The PDF is not merely a summary of slides. It is an organised narrative containing definitions, explanations, examples, interpretations, and mathematical arguments that can be examined more than once.

3. **Ask questions and clarify the material.** Mark every place where a definition, transition, calculation, or conclusion is not yet clear. Use the text, your own examples, classroom discussion, and appropriate digital tools to resolve these questions.

4. **Solve the complete assigned problem set.** Move from following an explanation to constructing arguments independently. Work on every assigned problem and prepare a separate, self-contained note for each one.

5. **Prepare professional mathematical notes.** Use precise notation, explanatory sentences, justified transitions, and a clear conclusion. The note should show not only the answer, but also how it follows from the given information, definitions, and earlier results.

6. **Present and discuss your work.** During exercise classes, be prepared to explain a solution, answer questions about its steps, and describe what you learned while preparing it.

These stages form one continuous process. Each stage prepares the next, and the presentation and feedback lead back to further reading, revision, and improved work.

Exercise Classes: Presentation, Feedback, and Revision

Presenting mathematics is part of learning mathematics. It reveals whether an argument has genuinely been understood and whether the student can communicate it clearly. During a presentation, the student should be able to read the notation, explain the role of each important formula, justify the method, and respond to questions without relying on AI to speak on the student's behalf.

Feedback given during exercise classes is intended for the whole group, not only for the student whose work is currently being presented. Everyone should follow the argument and listen carefully to the discussion. By comparing different examples, students learn what a complete solution looks like, which explanations make an argument clear, which steps require justification, and which weaknesses require revision.

Students should not focus only on comments addressed directly to their own work. Listening to feedback on the work of others provides a broader set of examples and develops an operational understanding of the course standards: what makes mathematical work clear, justified, and complete, and what still needs improvement.

The feedback concerns the mathematical work and its results, not the personal worth of the student. What is examined is the quality of the note, the understanding demonstrated in the argument, and the clarity of the presentation. Feedback should identify what already works and what should be changed in order to produce a stronger solution.

The first submitted version does not always have to be the final version. Students should learn to receive comments, reconsider their choices, correct errors, add missing explanations, and improve both the written note and its presentation. Revision is not evidence of failure. It is a normal part of serious intellectual work.

This process is important throughout university study and later in professional life. Students and professionals are given tasks, asked to solve problems, expected to present results, and required to respond to feedback and critical review. Learning to improve an initial solution is therefore not an additional classroom exercise. It is a fundamental habit of responsible academic and professional work.

Progress and Assessment

Progress is demonstrated by the quality and increasing independence of the student's work. By the end of a weekly cycle, the student should be able to identify the main mathematical objects, state the essential definitions, solve representative problems, justify the principal steps, verify or interpret the result, and communicate the argument clearly.

Assessment in the exercise component focuses primarily on how the student works throughout the semester. Evidence of serious engagement includes:

- regular preparation and completion of the assigned problem sets,
- documented attempts to solve problems rather than bare answers,
- active participation in presentations and classroom discussion,
- separate, self-contained notes for individual problems,
- careful revision of work after feedback,
- increasing clarity, independence, and mathematical precision.

A student is not defined by the quality of the first solution or presentation. Some students may initially find mathematical notation, formal writing, or public explanation difficult. A weak beginning does not prevent a very good final grade for the exercise component. A student who works conscientiously, responds to feedback, corrects earlier weaknesses, and demonstrates substantial development may finish the exercise component with a very good grade.

This approach does not lower the standard of mathematical work. It recognises that reaching the standard through sustained effort and revision is itself a meaningful achievement. The exercise-component grade reflects a process of learning, not merely a snapshot taken at the beginning of the semester.

Systematic work according to these guidelines provides the basis for earning a passing grade for the exercise component. The examination is assessed separately and tests the student's mathematical knowledge and ability to use it independently. Regular work does not replace the examination, but it develops the understanding and skills that substantially increase the likelihood of successfully completing the course.

How to Prepare Mathematical Notes

A mathematical note is not merely a place where an answer is recorded. Its purpose is to show how a problem is understood, how relevant information is selected, how a method is chosen, and how a conclusion follows from definitions, assumptions, and earlier results.

Mathematical expressions are compact. A single formula may contain several ideas at once: objects, operations, conditions, and logical relationships. Reading a formula therefore means more than pronouncing its symbols. A student should be able to explain what the objects represent, what operation is being performed, why the expression is meaningful, and what information the formula provides.

The same principle applies to writing. A sequence of formulas without explanatory sentences rarely communicates a complete mathematical argument. Formulas should be connected by words that explain their role. A good solution tells the reader why a calculation is being performed, which definition or theorem is being used, whether its assumptions are satisfied, and how the result answers the original question.

Make the Note Self-Contained

Prepare one separate, self-contained note for each assigned problem, including all of its subparts. Begin by writing the problem or a complete and faithful formulation of it. The reader should not need to search through another document to discover what is being solved. After several weeks, you should still be able to open the note and understand both the question and the solution.

Identify what is given and what must be found, proved, or explained. Introduce the relevant notation and determine what kinds of mathematical objects appear in the problem. Before calculating, ask what information is available and what would count as a satisfactory answer.

Make the Reasoning Visible

Present the principal idea or method before, or while, carrying out the calculation. Important transitions should be explained with complete sentences. For example:

- Because the determinant is non-zero, the matrix is invertible.
- We may therefore multiply both sides by the inverse matrix.
- By the definition of the derivative, we consider the following limit.
- Since the direction vector is non-zero, it determines a line.
- Substituting the result into the original equation verifies the solution.

Such sentences are not decoration around the mathematics. They are part of the mathematics because they show the logical relationship between successive formulas.

It is not necessary to describe every elementary arithmetic operation. The amount of explanation should be proportional to the problem. A simple exercise may require only a few sentences, while a more substantial problem may require a longer argument. The goal is not to make every note long. The goal is to make every essential step understandable and justified.

> Write as briefly as possible, but as completely as necessary.

A Complete Solution

A complete note should make the following five stages visible:

1. **Understand the problem.** Record the problem, identify the given information, and state clearly what must be found, proved, or explained.
2. **Choose a method.** Select the definitions, theorems, or techniques that connect the available information with the goal.
3. **Construct the argument.** Carry out the necessary steps and connect formulas with sentences that explain why each important transition is valid.

4. **Verify the result.** Check calculations, assumptions, domain restrictions, or geometric meaning whenever appropriate.

5. **State the conclusion.** Answer the original question clearly and in its proper context.

A result without its context is not a complete solution. A bare final answer, an unexplained sequence of formulas, or a few unconnected calculations will be treated as incomplete work and should be revised.

Quality cannot be replaced by quantity, but quality does not remove the obligation to complete the assigned work. Students are expected to prepare separate, complete solutions for the entire assigned problem set. Producing many unexplained answers is insufficient, but preparing one polished solution while ignoring the remaining problems is also insufficient.

Working with AI

We are studying in an age in which AI tools are widely available. The free versions of these tools are sufficient for the kinds of support expected in this course: asking questions, clarifying notation, generating simpler examples, checking reasoning, identifying missing steps, and improving the organisation of mathematical writing. Access to expensive software is not required.

AI should be used as an individual learning assistant. It may help transform rough calculations into clear mathematical prose, but it does not replace the student's mathematical work. Receiving a well-written answer is not the same as understanding it. The student remains responsible for checking every definition, calculation, assumption, conclusion, and formula included in the submitted note.

Use AI with a focused context. Work on one problem at a time and prepare one complete note for that problem. Do not paste an entire problem set into a single conversation and request all answers at once. Handling many unrelated tasks together encourages brief, generic, and mathematically weak notes. Focused work makes it easier to question each step, identify missing justifications, and produce a solution that can be understood and presented.

The one-problem-at-a-time principle does not mean that only one problem from the set should be prepared. Students are expected to repeat this focused process for every problem in the complete assigned set.

A polished AI-generated note is insufficient if the student cannot read its formulas, explain its terminology, justify its transitions, or present the argument independently. When using AI, ask not only for a solution, but also questions that deepen understanding:

- What does this formula mean?
- What mathematical object does each symbol represent?
- Why is this step valid?
- Which definition or theorem is being used?
- Are the assumptions of the theorem satisfied?
- Is there a simpler way to explain this transition?
- How can the result be checked?
- What should I be able to explain during class?

Preparing notes in this way is part of learning mathematics. It develops the ability to read notation, recognise mathematical objects, organise information, construct arguments, and communicate ideas precisely. Repeating this process for the complete problem set builds the habits of systematic and conscientious work that should accompany the student throughout university study and later professional life.

Student GitHub Workspace

The current instructions, problem templates, and technical workflow for student work are maintained in the public repository `dchorazkiewicz/Math_Problems_Repo`. Students should create a public fork of this repository and commit their notes to their own fork. The course repository is the common template; it is not the place where individual student solutions should be submitted.

The repository contains a short map of the working process:

- `README.md` explains the purpose and structure of the workspace and provides the weekly index;
- `SOURCE_MATERIAL.md` links the current lecture PDF and maps each week to the corresponding LaTeX problem source used by an AI assistant;
- `NOTE_GUIDELINES.md` defines the standard of a complete mathematical note;
- `AI_WORKFLOW.md` describes the one-problem-at-a-time workflow with AI, review, commits, presentation, feedback, and revision;
- `AGENTS.md` gives a connected AI assistant the operational rules for working safely in the student's fork;
- the `docs/` directory contains twelve weekly folders and one Markdown file for each problem.

The workspace uses Markdown for student notes, Git and GitHub for version history and public forks, MathJax for mathematical typesetting, MkDocs for the web presentation, and GitHub Actions for automatic validation and deployment. These instructions and templates may be updated independently of this PDF, so students should consult the repository for the current working procedure. The repository links back to this PDF and to its LaTeX sources. Students use the PDF as the readable course material containing the theory, definitions, examples, and problem sets. A connected AI assistant uses the current mapped LaTeX file as the exact machine-readable source of a selected problem statement.

Chapter 1

Cartesian Coordinates and Coordinate Systems

1.1 The Euclidean Geometry Available Before Coordinates

Before introducing coordinates, we recall the geometric world that is already available. At this stage there are no ordered pairs, no axes, and no vectors. There are points, straight lines, segments, circles, and angles, together with relations such as incidence, betweenness, congruence, perpendicularity, and parallelism.

Remark

Coordinates will later translate this geometry into arithmetic. They do not create the geometry. We can already construct and compare geometric objects before assigning numbers to all points and angles.

1.1.1 Three Different Levels Must Not Be Confused

We shall distinguish three kinds of statements.

1. *Euclid's postulates* describe the geometric world being assumed.
2. *Straightedge-and-compass rules* describe which elementary drawings and intersections may be used in a construction.
3. *Derived constructions* must be built from those elementary operations and justified by a proof.

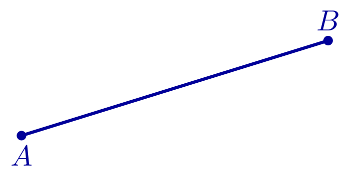
In particular, a perpendicular bisector, a perpendicular through a point, an angle bisector, or a parallel through a point will not be treated as primitive construction postulates. They will be constructed.

1.1.2 Euclid's Five Postulates

The following are the five postulates used in the first book of Euclid's *Elements*. The diagrams illustrate their meaning; they are not proofs.

Postulate I: Joining Two Points

A straight segment can be drawn from any point to any other point.



a segment is drawn from A to B

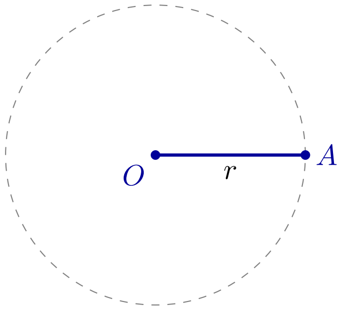
Postulate II: Extending a Segment

A finite straight segment can be extended continuously in the same straight line.

AB is extended beyond B

Postulate III: Drawing a Circle

A circle can be drawn with any chosen centre and any chosen radius.

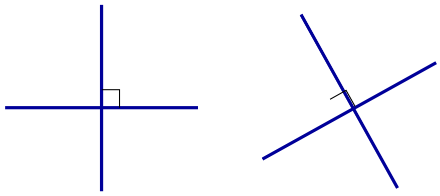


centre O and radius r determine a circle

This postulate is what permits a length to be copied geometrically without first assigning a number to it.

Postulate IV: Equality of Right Angles

All right angles are congruent.

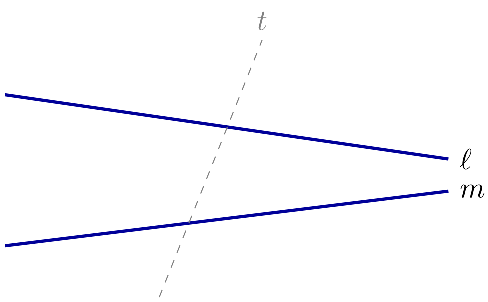


every right angle represents the same geometric magnitude

This postulate does not say that a perpendicular may simply be drawn without construction. It says that once right angles have been constructed, they are all equal.

Postulate V: The Parallel Postulate

If a transversal meets two straight lines and the interior angles on one side sum to less than two right angles, then the two lines, when extended, meet on that side.



the angle condition determines on which side ℓ and m meet

A familiar modern equivalent is Playfair's formulation: through a point outside a given line there passes exactly one parallel to that line. We shall use that form later, after the relation between the two formulations has been made clear.

1.1.3 Elementary Rules for Straightedge-and-Compass Work

For explicit constructions we use the following operational rules.

1. Through two already constructed distinct points, draw the straight line.
2. With an already constructed point as centre and an already constructed segment as radius, draw a circle.
3. Retain intersection points of two constructed lines, a line and a circle, or two circles whenever such intersections exist.

The first two rules are the operational forms of Euclid's first three postulates; the third explains how new constructible points are obtained. These rules are the complete elementary toolkit used in the constructions below.

1.1.4 Constructions Derived from the Toolkit

The Perpendicular Bisector and the Midpoint

Let A and B be distinct points. Draw two circles with the same radius, one centred at A and one centred at B, choosing the radius larger than half of $|AB|$. Let their intersection points be P and Q.

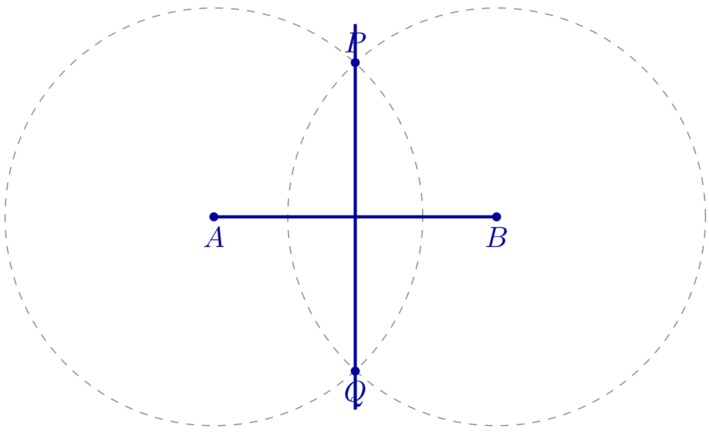
Because P lies on both circles,

$$|PA| = |PB|,$$

and similarly,

$$|QA| = |QB|.$$

Thus both P and Q are equidistant from A and B. The line PQ is the perpendicular bisector of AB.



The intersection of PQ with AB is the midpoint of AB. Thus midpoint and perpendicularity have been constructed from equal radii and intersections.

A Perpendicular Through a Point on a Line

Let P lie on a line ℓ.

Step 1. Draw a circle centred at P. It meets ℓ at two points A and B. Since A and B lie on the same circle,

$$|PA| = |PB|.$$

Thus P is the midpoint of AB.

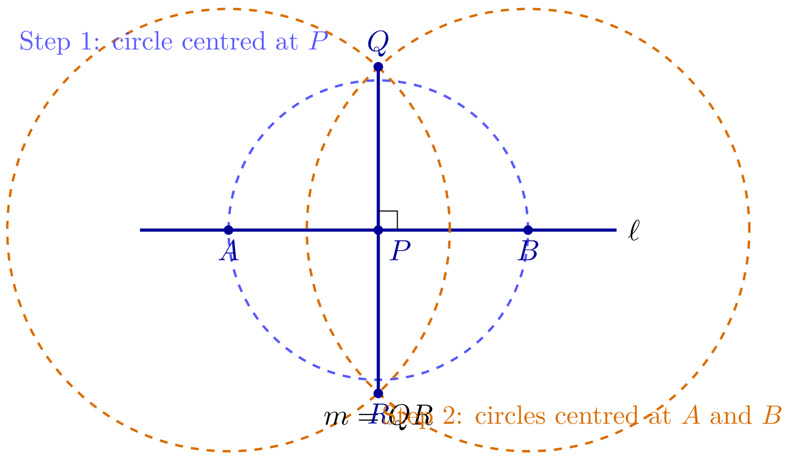
Step 2. Draw two circles with the same radius, one centred at A and one centred at B, choosing the radius larger than $|AP|$. Let their intersection points be Q and R. Then

$$|AQ| = |BQ| \qquad \text{and} \qquad |AR| = |BR|.$$

Both Q and R are therefore equidistant from A and B, so the line QR is the perpendicular bisector of AB. Because P is the midpoint of AB, the line QR passes through P. Since AB lies on ℓ,

$$QR \perp AB, \qquad \text{hence} \qquad QR \perp \ell.$$

The required perpendicular is the line $m = QR$.



The perpendicular has now been obtained entirely from the permitted straightedge-and-compass operations.

A Perpendicular Through a Point Outside a Line

Let P lie outside a line ℓ.

Step 1. Draw a circle centred at P with radius large enough to meet ℓ at two points A and B. Since A and B lie on the same circle,

$$|PA| = |PB|.$$

Step 2. Draw two circles of radius $|PA|$: one centred at A and one centred at B. The two circles meet at P and at a second point Q. Therefore

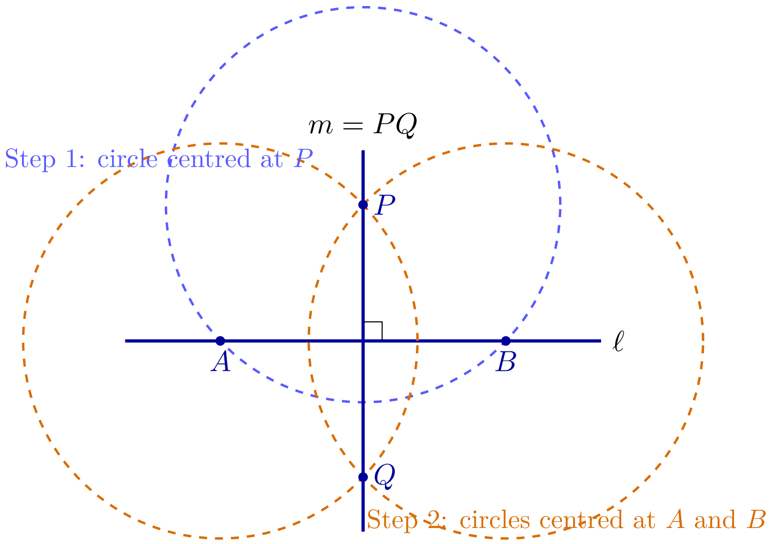
$$|AP| = |BP| \qquad \text{and} \qquad |AQ| = |BQ|.$$

Thus both P and Q are equidistant from A and B. The line PQ is therefore the perpendicular bisector of AB.

Because A and B lie on ℓ, the segment AB is contained in ℓ. Consequently,

$$PQ \perp AB, \qquad \text{hence} \qquad PQ \perp \ell.$$

The required perpendicular is the line $m = PQ$.



The perpendicular has now been obtained entirely from the permitted straightedge-and-compass operations: drawing circles, retaining their intersection points, and drawing the line through two constructed points.

Bisecting an Angle

Let two rays with common endpoint O form an angle. Draw a circle centred at O meeting the rays at A and B. Then draw equal-radius circles centred at A and B, and let one of their intersections inside the angle be P. Since

$$|OA| = |OB|, \qquad |AP| = |BP|, \qquad |OP| = |OP|,$$

the triangles OAP and OBP are congruent. Therefore

$$\angle AOP = \angle POB.$$

The ray OP is the angle bisector.

Constructing a Parallel Through a Point

Let P lie outside a line ℓ. First construct a line m through P perpendicular to ℓ. Then construct a line n through P perpendicular to m.

The transversal m forms equal corresponding right angles with n and ℓ. By the converse of the corresponding-angles theorem, the two lines are parallel:

$$n \parallel \ell.$$

The fifth postulate, equivalently Playfair's postulate, guarantees that this is the unique parallel to ℓ through P.

1.1.5 Parallel Lines and Proportional Segments

A central fact of Euclidean geometry is that parallel lines preserve ratios. This is the content of Thales' theorem in the form needed here.

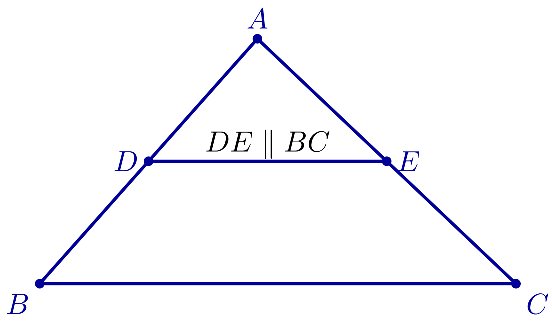
Theorem

Let D lie on AB and E lie on AC in a triangle ABC. If

$$DE \parallel BC,$$

then

$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}.$$



The smaller triangle ADE and the larger triangle ABC have the same three angles. Their corresponding sides differ by one common scale factor. This is the basic phenomenon of similarity.

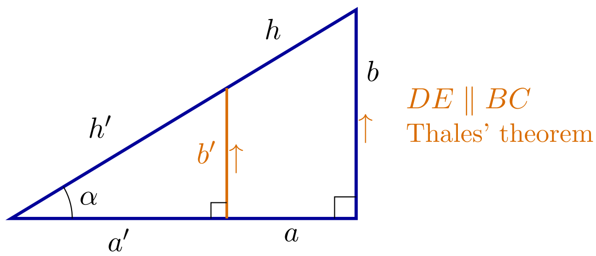
Definition

Two triangles are similar when their corresponding angles are equal. In that case their corresponding side lengths are proportional.

Similarity lets us change the size of a figure while preserving its shape. It is the reason ratios of corresponding lengths can depend only on angles.

1.1.6 Sine and Cosine Before Coordinates

Take a right triangle containing an acute angle α. Let h denote the hypotenuse, let a be the leg adjacent to α, and let b be the leg opposite α.



$$\frac{a'}{a} = \frac{b'}{b} = \frac{h'}{h}, \qquad \frac{a'}{h'} = \frac{a}{h}, \qquad \frac{b'}{h'} = \frac{b}{h}.$$

For an acute geometric angle α, define

$$\cos\alpha = \frac{a}{h}, \qquad \sin\alpha = \frac{b}{h}.$$

These ratios are independent of the chosen right triangle because all right triangles containing the same acute angle are similar.

For angles occurring in an arbitrary Euclidean triangle, we also need the right and obtuse cases. We set

$$\cos(\text{right angle}) = 0.$$

If α is obtuse and β is its acute supplement, so that together they form a straight angle, define

$$\cos\alpha = -\cos\beta.$$

Thus the geometric cosine is defined for every angle between 0 and a straight angle. For acute angles it is positive, for a right angle it is zero, and for obtuse angles it is negative.

The essential role of Thales' theorem and similarity is that these values depend only on the angle, not on the size of the triangle used to represent it.

For an acute angle, the Pythagorean theorem gives

$$a^2 + b^2 = h^2.$$

Dividing by h^2 yields

$$\left(\frac{a}{h}\right)^2 + \left(\frac{b}{h}\right)^2 = 1,$$

and hence

$$\cos^2\alpha + \sin^2\alpha = 1.$$

Thus the fundamental trigonometric identity is a direct reformulation of the Pythagorean theorem.

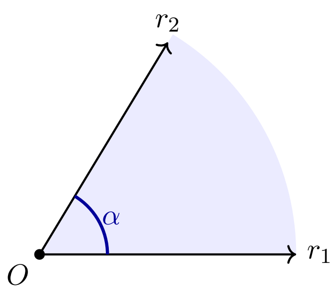
1.1.7 Angles and Their Numerical Measure

An angle exists geometrically before it is assigned a number. We can compare, copy, add, and bisect angles without degrees or radians. A numerical angle measure requires an additional choice of unit.

Definition

Let r_1 and r_2 be two rays with the same initial point O. The rays, together with a choice of one of the two regions between them, determine an angle α with vertex O.

The angle is the chosen geometric region. It is neither the small arc drawn near the vertex nor a number. The arc is only a graphical mark indicating which region has been selected.



Why circles give a consistent measure

Draw a circle centred at O. The rays cut from the circle an arc corresponding to the chosen angle.

Theorem

For every Euclidean circle, the ratio of circumference C to diameter D is the same constant:
$$\frac{C}{D} = \pi.$$
This constant is called π. Since $D = 2r$, every circle of radius r has circumference
$$C = 2\pi r.$$

The constancy of C/D follows from similarity: enlarging a circle by a factor λ multiplies every length, including its diameter, circumference, and corresponding arc lengths, by λ.

If the same two rays meet two circles of radii r_1 and r_2 and cut arc lengths s_1 and s_2, then
$$\frac{s_1}{s_2} = \frac{r_1}{r_2},$$
and therefore
$$\frac{s_1}{r_1} = \frac{s_2}{r_2}.$$
Thus the quotient s/r depends only on the angle, not on the chosen circle.

Definition

Let s be the length of the selected arc on a circle of radius r centred at the vertex. The radian measure of α is
$$m_{\text{rad}}(\alpha) = \frac{s}{r}.$$

For a full turn, $s = 2\pi r$, so
$$m_{\text{rad}}(\text{full turn}) = 2\pi.$$
A half-turn has radian measure π, and a quarter-turn has radian measure $\pi/2$.

Degrees are a second numerical scale. A full turn is assigned $360°$, so
$$2\pi \text{ radians} = 360°$$
and
$$\boxed{m_{\text{deg}}(\alpha) = \frac{180°}{\pi}\, m_{\text{rad}}(\alpha)}.$$

In particular, a constructible right angle has the numerical measures
$$m_{\text{rad}}(\text{right angle}) = \frac{\pi}{2}, \qquad m_{\text{deg}}(\text{right angle}) = 90°.$$
The angle is still the geometric region; $\pi/2$ and $90°$ are two numbers assigned to it by two different measurement systems.

Sine and cosine as functions of a number

The trigonometric ratios were first defined for acute geometric angles. We now extend them to a full turn. For $t \in [0, 2\pi)$, let $\beta \in [0, \pi/2]$ be the acute reference angle between the direction of t and the nearest horizontal ray. The magnitudes $\sin\beta$ and $\cos\beta$ come from the right-triangle definition; their signs are determined by the quadrant:

	$\cos t$	$\sin t$
$0 \le t \le \pi/2$	$+\cos\beta$	$+\sin\beta$
$\pi/2 \le t \le \pi$	$-\cos\beta$	$+\sin\beta$
$\pi \le t \le 3\pi/2$	$-\cos\beta$	$-\sin\beta$
$3\pi/2 \le t < 2\pi$	$+\cos\beta$	$-\sin\beta$

At the axes this gives the familiar values 0, 1, and -1. For arbitrary real t, define sine and cosine periodically:

$$\sin(t+2\pi k)=\sin t,\qquad \cos(t+2\pi k)=\cos t,\qquad k\in\mathbb{Z}.$$

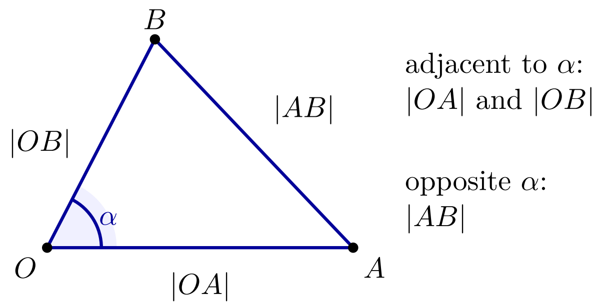
If $t=m_{\mathrm{rad}}(\alpha)$, then $\cos t$ is the same value as the geometric cosine $\cos\alpha$. This explains the notation $\cos(m_{\mathrm{rad}}(\alpha))$.

The Euclidean law of cosines

Directly measuring an arc is often difficult. A triangle provides another way to recover the measure of an angle. Choose points

$$A\in r_1\setminus\{O\},\qquad B\in r_2\setminus\{O\}.$$

The points form the Euclidean triangle OAB.



Theorem

If $\alpha=\angle AOB$, then the side lengths of the Euclidean triangle satisfy

$$|AB|^2=|OA|^2+|OB|^2-2|OA|\,|OB|\cos\alpha.$$

Equivalently, the last factor may be written as $\cos(m_{\mathrm{rad}}(\alpha))$.

The theorem belongs to Euclidean geometry; it does not require Cartesian coordinates. Solving it for the angle measure gives

$$\boxed{m_{\mathrm{rad}}(\alpha)=\arccos\left(\frac{|OA|^2+|OB|^2-|AB|^2}{2|OA|\,|OB|}\right)}.$$

Later, coordinates will provide a convenient arithmetic method for calculating the three Euclidean lengths $|OA|$, $|OB|$, and $|AB|$.

Remark

At this point no vector has been introduced. We have only Euclidean objects, postulates, constructions, similarity, trigonometric ratios, numerical angle measure, and the Euclidean law of cosines. Vectors will appear only in the next chapter, after the Cartesian coordinate system has been constructed.

The next step is to choose two perpendicular lines, an origin, directions, and a unit length. Those choices will convert the Euclidean plane into a Cartesian coordinate plane.

1.2 Constructing Cartesian Coordinates from Euclidean Geometry

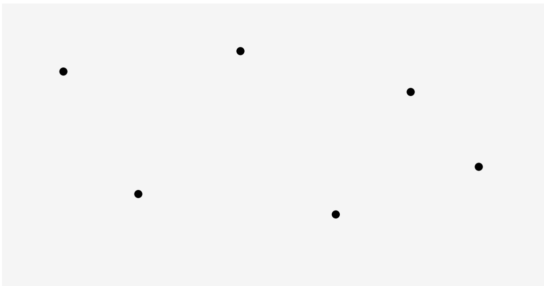
We begin with a Euclidean plane, not with ordered pairs and not with a coordinate system that has already been drawn. Points, straight lines, right angles, perpendicularity, congruent segments, and the Pythagorean theorem belong to the geometry available before coordinates are introduced. Our task is to construct, step by step, the additional choices that turn this plane into a Cartesian coordinate plane.

Remark

Coordinates do not create Euclidean geometry. They arise after we choose and construct a particular system for assigning numerical labels to points of an already existing Euclidean plane.

1.2.1 The Plane Before Coordinates

At first there is only a Euclidean plane, with no distinguished line, origin, directions, or numerical scale.



a Euclidean plane with no chosen coordinates

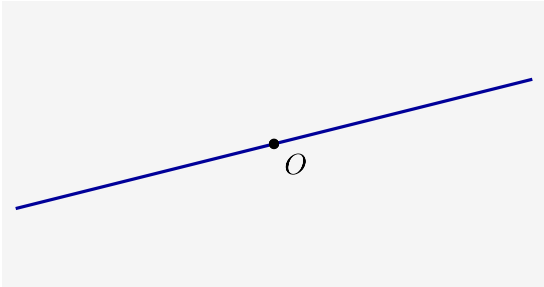
1.2.2 First Construction: A Straight Line

Choose two distinct points of the plane and construct the unique straight line through them. At this stage it is only a chosen Euclidean line. It is not yet an axis, because there is not yet a second line, an origin, a direction, or a scale.

the first constructed line

1.2.3 Second Construction: A Point on the Line

Choose an arbitrary point O on the constructed line. This point will later become the common zero of both coordinate axes, but it becomes an origin only after the rest of the coordinate structure has been constructed.

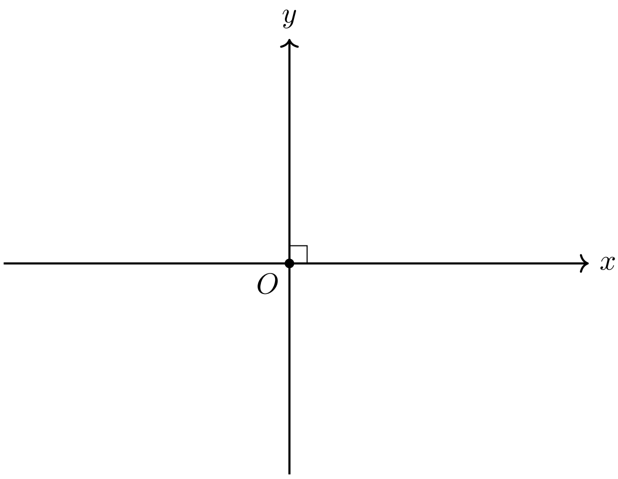


a point chosen on the first line

1.2.4 Third Construction: The Perpendicular Line

Through O, construct the unique line perpendicular to the first line. The two constructed lines now intersect at a right angle. Only now do we have the geometric framework from which two coordinate axes can be made.

We call the first line the x-axis, the perpendicular line the y-axis, and their intersection O the origin.

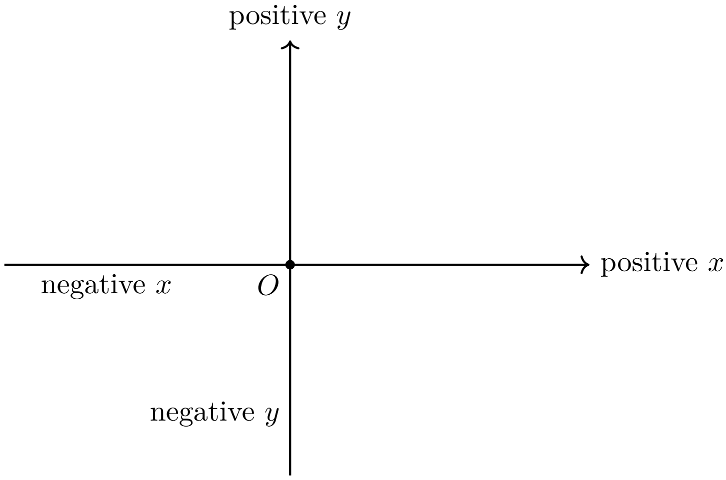


The axes are perpendicular because this was required in their geometric construction. No coordinate formula and no numerical angle measure has been used.

1.2.5 Choosing Positive and Negative Directions

Each axis is divided by O into two opposite rays. Choose one ray on the x-axis as positive and one ray on the y-axis as positive. The opposite rays are then called negative.

These choices are not forced by the Euclidean plane. Reversing one of them would produce a different coordinate system on the same geometric plane.

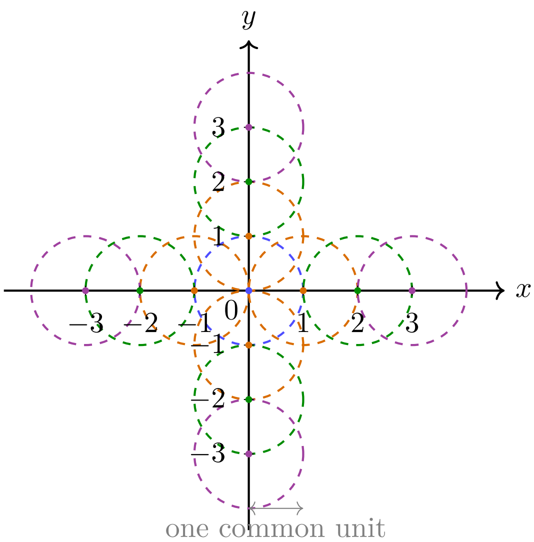


1.2.6 Constructing a Common Unit Length

Choose a positive segment length and call it one unit. Using a compass, transfer this same length from O repeatedly along both axes and in both directions. The first marks are labelled 1 and -1; repetition constructs the integer marks

$$\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$$

on each axis.



Remark

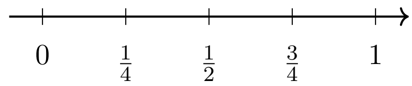
The same unit must be used on both axes. A segment labelled as one unit on the x-axis must be congruent to a segment labelled as one unit on the y-axis. This common scale is what later allows the Pythagorean theorem to produce the standard coordinate formula for Euclidean distance.

1.2.7 Refining the Scale by Bisection

The compass and straightedge construction does not stop at integer marks. Bisecting each unit segment constructs halves. Bisecting again constructs quarters, then eighths, and so on. Repeating this process on every integer interval and on both sides of the origin gives all dyadic marks

$$\frac{k}{2^n}, \qquad k \in \mathbb{Z}, \quad n \in \mathbb{N}.$$

successive bisections of one unit



The dyadic marks are dense on the axis: every interval, however small, contains such marks. They therefore provide arbitrarily fine numerical approximations to positions on the line.

Definition

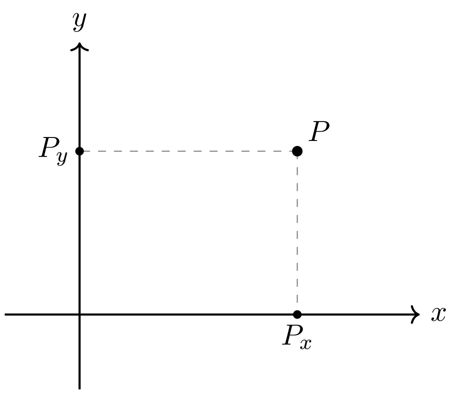
Continuity assumption. Once an origin, a positive direction, and a unit length are fixed on a Euclidean line, every point of the line corresponds to exactly one real number, and every real number corresponds to exactly one point of the line.

This assumption is additional to the finite straightedge-and-compass operations. Those operations construct many particular positions, including all dyadic marks, but not every real position. Continuity supplies the complete real scale used by coordinates. After this step, every point of each axis has a unique signed real label.

1.2.8 Assigning Coordinates by Orthogonal Projection

Only after the axes, directions, common unit, integer marks, refined scale, and real labels have been constructed can an arbitrary point of the plane be given coordinates.

Let P be any point of the plane. Through P, construct the line perpendicular to the x-axis. It meets the x-axis at a unique point P_x. Through P, construct the line perpendicular to the y-axis. It meets the y-axis at a unique point P_y.



If the signed real labels of P_x and P_y are x and y, respectively, we assign to P the ordered pair

$$P \longleftrightarrow (x, y).$$

The construction is unambiguous because the perpendicular through a given point to a given line is unique. Conversely, given real numbers x and y, locate the corresponding points on the two axes and construct through them lines perpendicular to their respective axes. Those two lines meet in exactly one point of the plane.

Theorem

Once the two perpendicular axes, their positive directions, and a common unit are fixed, orthogonal projection gives a one-to-one correspondence

$$\text{points of the Euclidean plane} \longleftrightarrow \text{pairs } (x, y) \in \mathbb{R}^2.$$

The ordered pair is the numerical address of a geometric point in the chosen coordinate system. It is not the point itself and it is not the source of the geometry.

1.2.9 The Coordinate Formula for Euclidean Distance

The Euclidean distance between two points is the length of the straight segment joining them. Coordinates now allow this geometric length to be calculated.

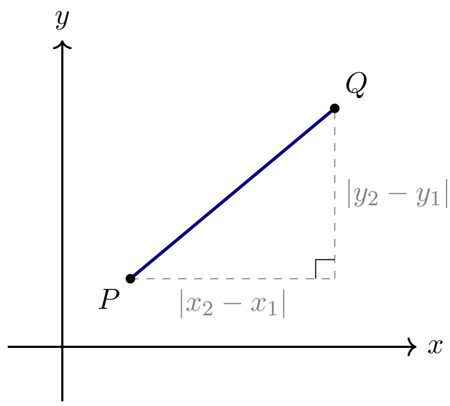
For points

$$P = (x_1, y_1), \qquad Q = (x_2, y_2),$$

their projections show that the horizontal and vertical separations have lengths

$$|x_2 - x_1| \qquad \text{and} \qquad |y_2 - y_1|.$$

Together with the segment PQ, these two perpendicular segments form a right triangle. Here "right" refers to the Euclidean right angle already used in the construction; no numerical measure of that angle is required.



By the Pythagorean theorem,

$$d_E(P, Q)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2,$$

and therefore

$$\boxed{d_E(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}.$$

The formula does not create a new notion of distance. It is the numerical coordinate expression of the Euclidean segment length already present in the plane.

Summary

The construction proceeds in the following order:

1. begin with a Euclidean plane containing no chosen coordinates;
2. construct one straight line;
3. choose a point O on that line;
4. construct through O the perpendicular line, and only then name the two lines as coordinate axes and O as their origin;
5. choose positive directions on both axes;
6. choose one common unit length and transfer it with a compass in both positive and negative directions to construct the integer marks;

7. repeatedly bisect intervals to construct the dense family of dyadic marks;
8. use Euclidean continuity to identify every axis position with a real number;
9. assign to every point its unique pair of orthogonal projections;
10. derive the coordinate distance formula from the resulting right triangle and the Pythagorean theorem.

1.3 Another Way to Describe a Point: Polar Coordinates

The Cartesian description of a point uses two signed coordinate values obtained by orthogonal projection onto the axes:

$$P = (x, y).$$

This is natural when horizontal and vertical directions are important. It is not, however, the only possible way to describe position in the plane. We may instead ask two different questions:

1. How far is the point from the origin?
2. In which direction do we move from the origin to reach it?

These questions lead to *polar coordinates*. For a point different from the origin, we use a pair

$$(r, \theta),$$

where

$$r > 0, \qquad 0 \le \theta < 2\pi.$$

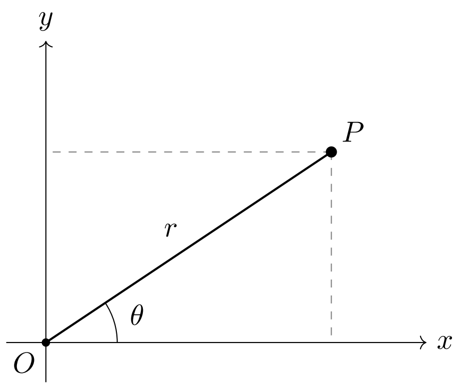
The number r is the Euclidean distance from the origin, and θ is the angle measured counterclockwise from the positive x-axis. The interval for the angle is half-open: it contains 0, but it does not contain 2π. This choice prevents the same direction from being written twice.

The origin is handled separately. In the usual non-unique polar notation every pair $(0, \theta)$ represents the origin. In this course we choose the single representative

$$O = (0, 0)$$

so that every point has exactly one allowed polar pair. Thus the set of coordinate pairs used here is

$$\mathcal{P} = \{(0, 0)\} \cup \big((0, \infty) \times [0, 2\pi)\big).$$



The Cartesian and polar descriptions refer to the same point. The right triangle in the figure gives

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

Conversely, if $(x, y) \neq (0, 0)$, then

$$r = \sqrt{x^2 + y^2} > 0.$$

The polar angle is the unique number $\theta \in [0, 2\pi)$ satisfying

$$\boxed{\cos\theta = \frac{x}{r}, \qquad \sin\theta = \frac{y}{r}}.$$

These two equations determine both the quadrant and the exact direction. The single formula $\arctan(y/x)$ is insufficient because it loses quadrant information and is undefined when $x = 0$. In computation this unique angle is usually obtained with the two-argument function $\mathrm{atan2}(y, x)$, with a negative output shifted by 2π.

Example

Consider the point whose polar coordinates are

$$(r, \theta) = \left(2, \frac{\pi}{3}\right).$$

Its Cartesian coordinates are

$$x = 2\cos\frac{\pi}{3} = 1, \qquad y = 2\sin\frac{\pi}{3} = \sqrt{3}.$$

Thus the same point may be written as

$$P = \left(1, \sqrt{3}\right)$$

in Cartesian coordinates and as

$$P = \left(2, \frac{\pi}{3}\right)$$

in polar coordinates.

Polar coordinates are especially convenient when distance from the origin is the main geometric feature. Let

$$C_R = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = R^2\}, \qquad R > 0.$$

This is the set of all points in the Cartesian plane whose coordinates satisfy

$$x^2 + y^2 = R^2.$$

Geometrically, C_R is the circle of radius R centred at the origin. After substituting

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

the Cartesian condition becomes

$$r^2\cos^2\theta + r^2\sin^2\theta = R^2.$$

Since $\cos^2\theta + \sin^2\theta = 1$, this reduces to

$$\boxed{r = R}.$$

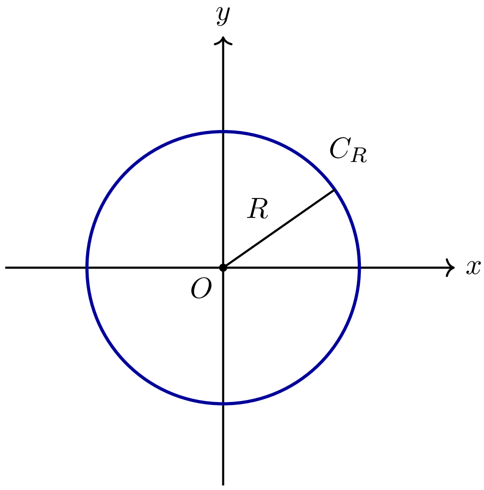
In polar coordinates, the same circle is therefore represented by the set

$$\{(R,\theta) : 0 \le \theta < 2\pi\}.$$

The set C_R has not changed. The conditions

$$x^2 + y^2 = R^2 \qquad \text{and} \qquad r = R, \quad 0 \le \theta < 2\pi,$$

describe the same geometric subset of the plane, but they use different coordinates. In Cartesian coordinates we express a relation between horizontal and vertical position. In polar coordinates we state directly that every point of the set has the same distance from the origin.



Remark

The restriction $0 \le \theta < 2\pi$ is part of the definition used here. The angle 2π is not included because it has the same direction as 0. We choose one representative angle for each direction rather than identifying the two endpoints of the interval.

Chapter 2

Vectors in the Plane and in Space

2.1 Vectors from Coordinate Displacements

In the previous chapter a point of the Euclidean plane was identified with an ordered pair of real numbers. We now use that identification immediately. The aim is not to collect formulas about vectors, but to let the formulas emerge from geometric questions and coordinate calculations.

Choose two points

$$A = (1, 2), \qquad B = (4, 6).$$

To pass from A to B, the first coordinate changes by 3 and the second by 4. The pair

$$(3, 4)$$

records the complete displacement.

2.1.1 A Vector Records a Change of Coordinates

For arbitrary points

$$A = (x_1, y_1), \qquad B = (x_2, y_2),$$

the displacement from A to B is

$$\overrightarrow{AB} = (x_2 - x_1,\ y_2 - y_1).$$

Definition

A vector in the Cartesian plane is an ordered pair

$$\mathbf{v} = (v_1, v_2)$$

interpreted as a coordinate displacement: v_1 units horizontally and v_2 units vertically.

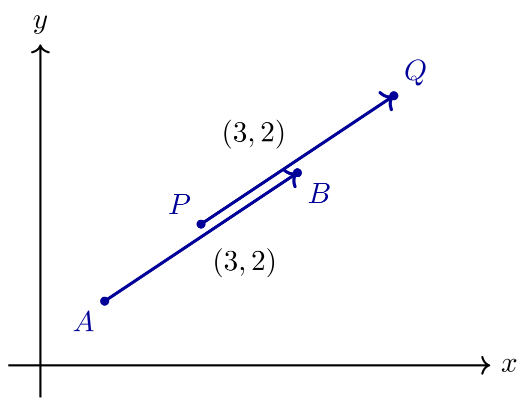
This makes the phrase "the same vector starting elsewhere" precise. If $P = (p_1, p_2)$ and

$$Q = (p_1 + v_1,\ p_2 + v_2),$$

then

$$\overrightarrow{PQ} = (v_1, v_2) = \mathbf{v}.$$

The same two coordinate changes have simply been applied at another point.



The zero displacement is

$$\mathbf{0} = (0, 0).$$

Reversing $\mathbf{v} = (v_1, v_2)$ gives

$$-\mathbf{v} = (-v_1, -v_2).$$

2.1.2 Two Successive Displacements Force Vector Addition

Start at an arbitrary point (x, y). First apply

$$\mathbf{u} = (u_1, u_2),$$

and then

$$\mathbf{v} = (v_1, v_2).$$

The successive endpoints are

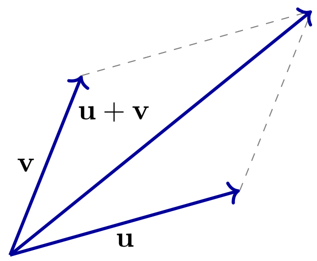
$$(x, y) \longmapsto (x + u_1, y + u_2) \longmapsto (x + u_1 + v_1, y + u_2 + v_2).$$

Therefore the single displacement with the same effect must be

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1,\ u_2 + v_2).$$

The rule is forced by the experiment; it is not introduced as a convention to memorise.

Because addition of real numbers is commutative, applying $\mathbf{u}$ and then $\mathbf{v}$ reaches the same endpoint as applying $\mathbf{v}$ and then $\mathbf{u}$. This produces the parallelogram rule.



Subtraction is the displacement that undoes another displacement:

$$\mathbf{u} - \mathbf{v} = (u_1 - v_1,\ u_2 - v_2).$$

In particular,

$$\overrightarrow{AB} = B - A.$$

2.1.3 Repeating and Reversing a Displacement

Repeating $\mathbf{v} = (v_1, v_2)$ twice gives $(2v_1, 2v_2)$, while reversing it gives $(-v_1, -v_2)$. The natural extension is

$$\lambda\mathbf{v} = (\lambda v_1, \lambda v_2), \qquad \lambda \in \mathbb{R}.$$

For positive λ the orientation is unchanged; for negative λ it is reversed.

2.1.4 Changing the Basis: New Coordinates for the Same Vector

A vector is a geometric displacement. Its coordinate column depends on the chosen basis. Changing the basis does not rotate or deform the vector itself; it changes only the pair of numbers used to describe it.

What a basis is

Definition

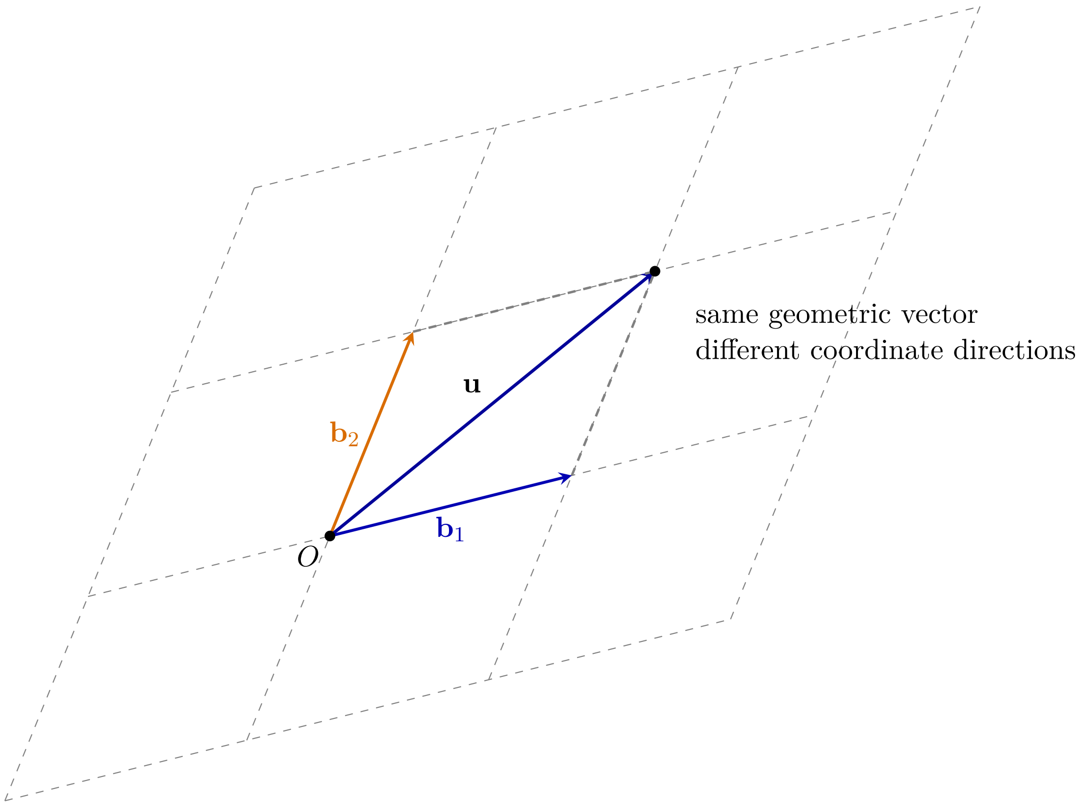
Two vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ form a basis of the plane when they are not parallel. Equivalently, every vector $\mathbf{u}$ can then be written in one and only one way as

$$\mathbf{u} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2.$$

The coordinate column of $\mathbf{u}$ in this basis is

$$[\mathbf{u}]_{\mathcal{B}} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}, \qquad \mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2).$$

The vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ need not be perpendicular and need not have unit length. They only need to point in different directions.



The dashed parallelogram shows how the endpoint of $\mathbf{u}$ is reached by moving first in the $\mathbf{b}_1$ direction and then in the $\mathbf{b}_2$ direction. These two signed amounts are the coordinates c_1 and c_2.

The standard basis and a slanted basis

The standard basis is

$$\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2), \qquad \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

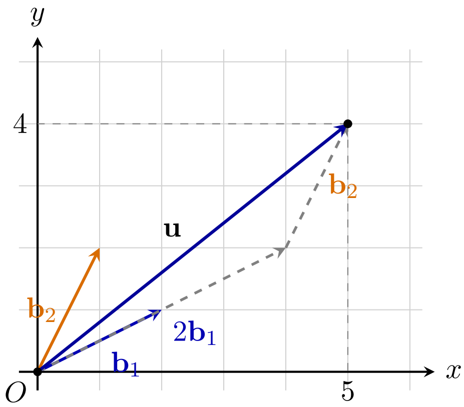
For this basis,

$$\mathbf{u} = x\mathbf{e}_1 + y\mathbf{e}_2 \quad \iff \quad [\mathbf{u}]_{\mathcal{E}} = \begin{pmatrix} x \\ y \end{pmatrix}.$$

Now choose the slanted basis

$$\mathbf{b}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The same vector
$$\mathbf{u} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$
has standard coordinates $(5, 4)$, but in the slanted basis we solve
$$\begin{pmatrix} 5 \\ 4 \end{pmatrix} = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$
This gives
$$2c_1 + c_2 = 5, \qquad c_1 + 2c_2 = 4,$$
so
$$c_1 = 2, \qquad c_2 = 1.$$
Therefore
$$\boxed{[\mathbf{u}]_{\mathcal{E}} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}, \qquad [\mathbf{u}]_{\mathcal{B}} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}}.$$



The vector has not changed. Only the coordinate grid used to count its components has changed from rectangular to slanted.

The basis matrix

Place the basis vectors as columns of a matrix:
$$B = \begin{pmatrix} | & | \\ \mathbf{b}_1 & \mathbf{b}_2 \\ | & | \end{pmatrix}.$$
Then the equation
$$\mathbf{u} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2$$
becomes
$$\boxed{[\mathbf{u}]_{\mathcal{E}} = B[\mathbf{u}]_{\mathcal{B}}}.$$
Because $\mathbf{b}_1$ and $\mathbf{b}_2$ are not parallel, B is invertible. Hence
$$\boxed{[\mathbf{u}]_{\mathcal{B}} = B^{-1}[\mathbf{u}]_{\mathcal{E}}}.$$

For the example,
$$B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad B^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$
Therefore
$$[\mathbf{u}]_{\mathcal{B}} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Changing directly from one basis to another

Let

$$\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2), \qquad \mathcal{C} = (\mathbf{c}_1, \mathbf{c}_2),$$

and let B and C be their basis matrices in standard coordinates. Since

$$[\mathbf{u}]_{\mathcal{E}} = B[\mathbf{u}]_{\mathcal{B}} = C[\mathbf{u}]_{\mathcal{C}},$$

we obtain

$$\boxed{[\mathbf{u}]_{\mathcal{C}} = C^{-1}B[\mathbf{u}]_{\mathcal{B}}}.$$

The matrix

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = C^{-1}B$$

is the change-of-basis matrix from $\mathcal{B}$-coordinates to $\mathcal{C}$-coordinates.

Remark

The columns of $P_{\mathcal{C}\leftarrow\mathcal{B}}$ are the $\mathcal{C}$-coordinates of the old basis vectors $\mathbf{b}_1$ and $\mathbf{b}_2$. This gives a direct geometric interpretation of every column.

Why ordinary perpendicular projections fail for a slanted basis

For an orthonormal basis $(\mathbf{e}_1, \mathbf{e}_2)$, the coordinates are

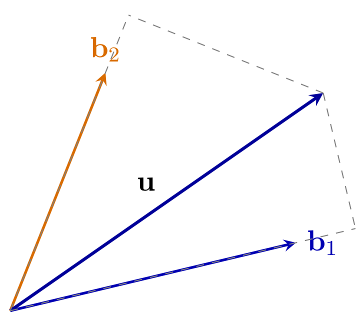
$$x = \mathbf{u} \cdot \mathbf{e}_1, \qquad y = \mathbf{u} \cdot \mathbf{e}_2.$$

This works because the basis vectors are perpendicular unit vectors.

For a slanted basis, however,

$$c_1 \neq \frac{\mathbf{u} \cdot \mathbf{b}_1}{\mathbf{b}_1 \cdot \mathbf{b}_1} \quad \text{and generally} \quad c_2 \neq \frac{\mathbf{u} \cdot \mathbf{b}_2}{\mathbf{b}_2 \cdot \mathbf{b}_2}.$$

Those expressions are perpendicular projections onto the two basis directions, but the two projected pieces generally do not add back to $\mathbf{u}$.



perpendicular projections do not form the oblique parallelogram components

There is nevertheless a projection-like way to read the coefficients. One uses the *dual basis* $(\mathbf{b}^1, \mathbf{b}^2)$, defined by

$$\mathbf{b}^i \cdot \mathbf{b}_j = \delta^i_j.$$

Then

$$\boxed{c_1 = \mathbf{b}^1 \cdot \mathbf{u}, \qquad c_2 = \mathbf{b}^2 \cdot \mathbf{u}}.$$

Thus ordinary dot products extract coordinates only after the correct dual directions have been constructed. Algebraically, the rows of B^{-1} are the transpose coordinates of the dual basis vectors.

Summary

A change of basis changes coordinates, not the geometric vector:

$$\mathbf{u} \quad \text{is fixed, while} \quad [\mathbf{u}]_{\mathcal{B}} \quad \text{depends on } \mathcal{B}.$$

For a basis matrix B,

$$[\mathbf{u}]_{\mathcal{E}} = B[\mathbf{u}]_{\mathcal{B}}, \qquad [\mathbf{u}]_{\mathcal{B}} = B^{-1}[\mathbf{u}]_{\mathcal{E}}.$$

Rectangular coordinates are a special case coming from an orthonormal basis. Slanted coordinates use the same principle, but their coefficients are found from a linear system rather than from ordinary perpendicular projections onto the basis vectors.

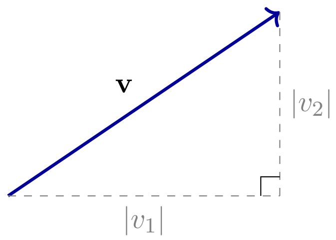
2.1.5 Length and Inclination

Draw $\mathbf{v} = (v_1, v_2)$ from the origin. Its endpoint and the coordinate axes form a right triangle. The Pythagorean theorem gives

$$\|\mathbf{v}\|^2 = v_1^2 + v_2^2, \qquad \|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}.$$

Consequently,

$$d(A, B) = \|B - A\|.$$



Let α be the angle of inclination of a nonzero vector $\mathbf{v}$ from the positive horizontal direction. The coordinate projections of $\mathbf{v}$ are

$$v_1 = \|\mathbf{v}\| \cos\alpha, \qquad v_2 = \|\mathbf{v}\| \sin\alpha.$$

Thus

$$\mathbf{v} = \|\mathbf{v}\|(\cos\alpha, \sin\alpha).$$

This representation applies to every nonzero vector; no vector has been chosen as a special coordinate axis.

2.1.6 One Angle Calculation, Two Geometric Tests

Take two arbitrary nonzero vectors

$$\mathbf{u} = (u_1, u_2), \qquad \mathbf{v} = (v_1, v_2).$$

Let their angles of inclination be α and β. Then

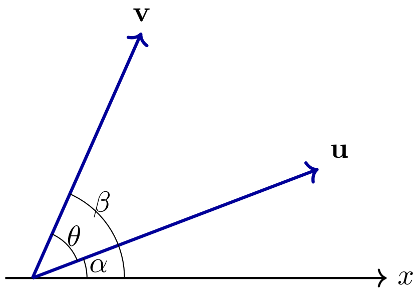
$$\mathbf{u} = \|\mathbf{u}\|(\cos\alpha, \sin\alpha),$$

$$\mathbf{v} = \|\mathbf{v}\|(\cos\beta, \sin\beta).$$

The difference $\beta - \alpha$ measures the directed turn from $\mathbf{u}$ to $\mathbf{v}$. The ordinary angle between the vectors will be denoted by

$$\theta = |\beta - \alpha|$$

when the smaller angle is intended.



Parallel directions arise from the sine of the angle difference

Two nonzero vectors determine the same unoriented direction precisely when the turn from one to the other is an integer multiple of a half-turn:

$$\beta - \alpha \equiv 0 \pmod{\pi}.$$

Equivalently,

$$\sin(\beta - \alpha) = 0.$$

Using the sine-difference identity,

$$\sin(\beta - \alpha) = \sin\beta\cos\alpha - \cos\beta\sin\alpha.$$

Now substitute

$$\cos\alpha = \frac{u_1}{\|\mathbf{u}\|}, \qquad \sin\alpha = \frac{u_2}{\|\mathbf{u}\|},$$
$$\cos\beta = \frac{v_1}{\|\mathbf{v}\|}, \qquad \sin\beta = \frac{v_2}{\|\mathbf{v}\|}.$$

We obtain

$$\begin{aligned}\sin(\beta - \alpha) &= \frac{v_2}{\|\mathbf{v}\|}\frac{u_1}{\|\mathbf{u}\|} - \frac{v_1}{\|\mathbf{v}\|}\frac{u_2}{\|\mathbf{u}\|} \\ &= \frac{u_1v_2 - u_2v_1}{\|\mathbf{u}\|\,\|\mathbf{v}\|}.\end{aligned}$$

The denominator is nonzero. Therefore

$$\sin(\beta - \alpha) = 0 \iff u_1v_2 - u_2v_1 = 0.$$

We have discovered the coordinate test

$$\mathbf{u} \parallel \mathbf{v} \iff u_1v_2 - u_2v_1 = 0.$$

The same condition implies that one vector is a scalar multiple of the other. Indeed, if $v_1 \neq 0$, set $\lambda = u_1/v_1$. The equality

$$u_1v_2 - u_2v_1 = 0$$

gives $u_2 = \lambda v_2$, and hence

$$\mathbf{u} = \lambda\mathbf{v}.$$

If $v_1 = 0$, then $v_2 \neq 0$ and the same argument uses $\lambda = u_2/v_2$. Thus

$$\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} = \lambda\mathbf{v} \quad \text{for some } \lambda \neq 0.$$

This scalar-multiple criterion is a conclusion of the angle calculation, not a rule imposed in advance.

The cosine of the same difference reveals perpendicularity

The vectors are perpendicular precisely when their angle difference is a quarter-turn modulo a half-turn:

$$\beta - \alpha \equiv \frac{\pi}{2} \pmod{\pi}.$$

Equivalently,

$$\cos(\beta - \alpha) = 0.$$

The cosine-difference identity gives

$$\cos(\beta - \alpha) = \cos\beta\cos\alpha + \sin\beta\sin\alpha.$$

Substituting the coordinate ratios gives

$$\begin{aligned}\cos(\beta - \alpha) &= \frac{v_1}{\|\mathbf{v}\|}\frac{u_1}{\|\mathbf{u}\|} + \frac{v_2}{\|\mathbf{v}\|}\frac{u_2}{\|\mathbf{u}\|} \\ &= \frac{u_1v_1 + u_2v_2}{\|\mathbf{u}\|\,\|\mathbf{v}\|}.\end{aligned}$$

Since the denominator is nonzero,

$$\mathbf{u} \perp \mathbf{v} \iff u_1v_1 + u_2v_2 = 0.$$

The coordinate expression has emerged from the geometry. Only now do we give it a name.

Definition

For vectors in the plane, their *dot product* is

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2.$$

In three-dimensional space,

$$(u_1, u_2, u_3) \cdot (v_1, v_2, v_3) = u_1v_1 + u_2v_2 + u_3v_3.$$

The preceding calculation proves

$$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0.$$

It proves more than the right-angle case. Because cosine is even and $\theta = |\beta - \alpha|$,

$$\cos(\beta - \alpha) = \cos\theta.$$

Therefore

$$\boxed{\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta}$$

for every pair of nonzero vectors in the plane.

Example

Take

$$\mathbf{u} = (3, 2), \qquad \mathbf{v} = (-2, 3).$$

Then

$$\mathbf{u} \cdot \mathbf{v} = 3(-2) + 2(3) = 0.$$

The coordinate calculation says that

$$\cos\theta = 0,$$

so the angle between the vectors is a right angle.

2.1.7 The Cosine Theorem Emerges Immediately

Draw $\mathbf{u}$ and $\mathbf{v}$ from a common point. The displacement from the endpoint of $\mathbf{v}$ to the endpoint of $\mathbf{u}$ is

$$\mathbf{u}-\mathbf{v}.$$

Thus these three vectors determine a triangle with side lengths

$$\|\mathbf{u}\|, \qquad \|\mathbf{v}\|, \qquad \|\mathbf{u}-\mathbf{v}\|.$$

We calculate the square of the third side:

$$\begin{aligned}
\|\mathbf{u}-\mathbf{v}\|^2 &= (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) \\
&= \mathbf{u}\cdot\mathbf{u}-\mathbf{u}\cdot\mathbf{v}-\mathbf{v}\cdot\mathbf{u}+\mathbf{v}\cdot\mathbf{v} \\
&= \|\mathbf{u}\|^2+\|\mathbf{v}\|^2-2\mathbf{u}\cdot\mathbf{v} \\
&= \|\mathbf{u}\|^2+\|\mathbf{v}\|^2-2\|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta.
\end{aligned}$$

If

$$a=\|\mathbf{u}\|, \qquad b=\|\mathbf{v}\|, \qquad c=\|\mathbf{u}-\mathbf{v}\|,$$

then

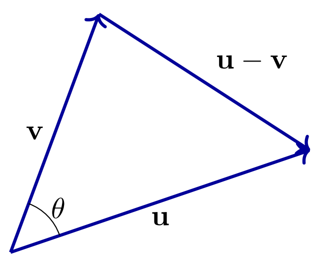
$$\boxed{c^2=a^2+b^2-2ab\cos\theta}.$$

Theorem

For a triangle with side lengths a, b, and c, where θ is the angle between the sides of lengths a and b,

$$c^2=a^2+b^2-2ab\cos\theta.$$

The cosine theorem is not inserted as an external formula. It is the coordinate expansion of the displacement between the endpoints of two arbitrary vectors.



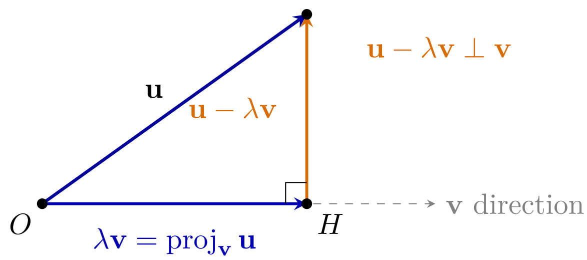
2.1.8 Projection Is Forced by the Right-Angle Condition

Let $\mathbf{v}\neq\mathbf{0}$. We seek a multiple $\lambda\mathbf{v}$ such that the remainder

$$\mathbf{u}-\lambda\mathbf{v}$$

is perpendicular to $\mathbf{v}$. Geometrically, this means decomposing $\mathbf{u}$ into a component parallel to $\mathbf{v}$ and a component perpendicular to $\mathbf{v}$:

$$\mathbf{u}=\lambda\mathbf{v}+(\mathbf{u}-\lambda\mathbf{v}).$$



The point H is chosen on the line spanned by $\mathbf{v}$. The vector from O to H is the parallel component $\lambda\mathbf{v}$, while the vector from H to the endpoint of $\mathbf{u}$ is the perpendicular remainder. The right-angle condition determines the value of λ.

The criterion just derived requires

$$(\mathbf{u} - \lambda\mathbf{v}) \cdot \mathbf{v} = 0.$$

Expanding gives

$$\mathbf{u} \cdot \mathbf{v} - \lambda(\mathbf{v} \cdot \mathbf{v}) = 0,$$

so

$$\lambda = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}.$$

Thus the parallel component is

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

The formula is the solution of a geometric requirement, not a rule to be memorised without explanation.

2.1.9 The Same Coordinate Ideas in Three Dimensions

In $\mathbb{R}^3$, a vector is a displacement

$$\mathbf{v} = (v_1, v_2, v_3),$$

and addition and scaling remain componentwise. Applying the Pythagorean theorem twice gives

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

Two nonparallel vectors in space determine two independent directions. To construct a vector perpendicular to both, solve the corresponding two dot-product conditions. The resulting expression is

$$\mathbf{u} \times \mathbf{v} = \big(u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1\big).$$

Direct substitution verifies

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0, \qquad (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0.$$

Thus the cross product is introduced as a constructed solution to two explicit perpendicularity conditions.

Summary

A vector was introduced as a coordinate displacement. Successive displacements forced vector addition, and repetition forced scalar multiplication. Writing arbitrary vectors through their lengths and inclination angles allowed the sine of the angle difference to produce the parallelism test and the cosine of the same difference to produce the dot product, perpendicularity, and the formula

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos \theta.$$

Expanding $\|\mathbf{u} - \mathbf{v}\|^2$ then produced the cosine theorem. Projection and the cross product arose by solving explicit perpendicularity conditions.

Chapter 3

Lines and Planes in the Plane and in Space

3.1 Lines in the Cartesian Plane

A line is one of the simplest geometric subsets of the plane. It can be drawn, described by a condition on coordinates, or defined explicitly as a subset of $\mathbb{R}^2$. We will keep these descriptions separate and explain how they are connected.

3.1.1 A Line as a Set of Points

Consider the condition

$$y = 2x + 1.$$

This is a relation between the coordinates x and y. The corresponding geometric object is the set

$$L = \{(x, y) \in \mathbb{R}^2 : y = 2x + 1\}.$$

Definition

A non-vertical line in the Cartesian plane is a set of the form

$$L_{m,b} = \{(x, y) \in \mathbb{R}^2 : y = mx + b\},$$

where $m, b \in \mathbb{R}$ are fixed.

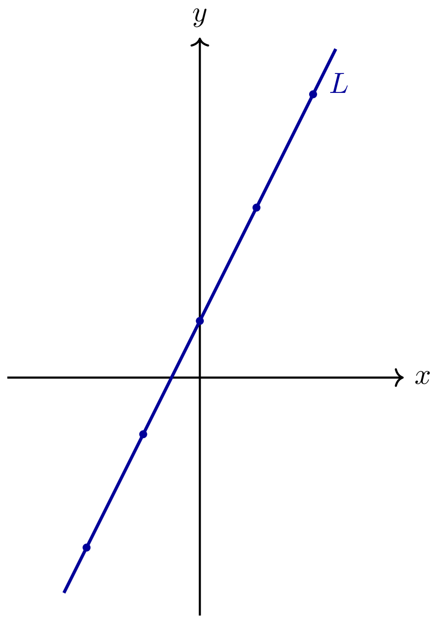
Every value of x determines exactly one value of y. For example, for $y = 2x + 1$ we obtain

x	-2	-1	0	1	2
y	-3	-1	1	3	5

and therefore all of the points

$$(-2, -3), \quad (-1, -1), \quad (0, 1), \quad (1, 3), \quad (2, 5)$$

belong to L.



Remark

A finite list of points only gives examples of elements of a line. The complete line is the entire set of points satisfying its defining condition.

3.1.2 Slope, Angle, and Vertical Change

Take two distinct points on a non-vertical line. Let their horizontal and vertical changes be

$$\Delta x = x_2 - x_1, \qquad \Delta y = y_2 - y_1.$$

Because the line is not vertical, $\Delta x \neq 0$.

Definition

The *slope* of a non-vertical line is

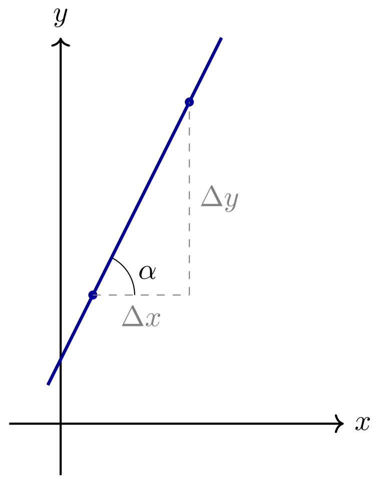
$$m = \frac{\Delta y}{\Delta x}.$$

It measures vertical change per unit of horizontal change.

For a line described by $y = mx + b$, replacing x by $x + \Delta x$ changes y by

$$\Delta y = m\,\Delta x.$$

Thus the coefficient m in the equation is exactly the geometric slope.



The line and the positive horizontal direction form an angle α. In the right triangle shown above, the tangent of α is defined as

$$\tan\alpha = \frac{\text{opposite side}}{\text{adjacent side}} = \frac{\Delta y}{\Delta x}.$$

Theorem

If α is an angle of inclination of a non-vertical line, then

$$m = \tan\alpha.$$

Thus slope is both a coordinate ratio and a description of direction.

The sign of m has a direct geometric meaning:

- $m > 0$: the line rises from left to right;
- $m < 0$: the line falls from left to right;
- $m = 0$: the line is horizontal.

3.1.3 The Constant Term and Axis Intersections

For the line

$$y = mx + b,$$

setting $x = 0$ gives $y = b$.

Theorem

The line

$$L_{m,b} = \{(x, y) \in \mathbb{R}^2 : y = mx + b\}$$

meets the y-axis at

$$(0, b).$$

To find an intersection with the x-axis, set $y = 0$ and solve the resulting equation for x.

Example

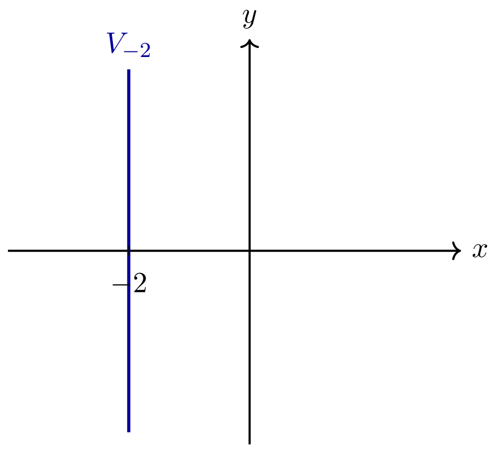
For
$$L = \{(x, y) \in \mathbb{R}^2 : y = -\tfrac{1}{2}x + 3\},$$
the y-axis intersection is $(0, 3)$. Setting $y = 0$ gives
$$0 = -\frac{1}{2}x + 3,$$
so $x = 6$. Hence the x-axis intersection is $(6, 0)$.

3.1.4 Vertical Lines

A vertical line cannot be written as $y = mx + b$, because one value of x is paired with every real value of y.

Definition

For a fixed $c \in \mathbb{R}$, the vertical line through $x = c$ is
$$V_c = \{(x, y) \in \mathbb{R}^2 : x = c\}.$$



Remark

A vertical line has no real-valued slope. For two distinct points on it, $\Delta x = 0$, so the quotient $\Delta y/\Delta x$ is undefined.

3.1.5 The General Linear Equation

Definition

Let $a, b, c \in \mathbb{R}$, where a and b are not both zero. The set
$$L_{a,b,c} = \{(x, y) \in \mathbb{R}^2 : ax + by = c\}$$
is a line in the Cartesian plane.

If $b \neq 0$, solving for y gives
$$y = -\frac{a}{b}x + \frac{c}{b}.$$

Therefore

$$\boxed{m = -\frac{a}{b}}, \qquad \boxed{b_{\text{intercept}} = \frac{c}{b}}.$$

If $b = 0$, then $a \neq 0$ and the condition becomes

$$x = \frac{c}{a},$$

which describes a vertical line.

Example

For

$$L = \{(x, y) \in \mathbb{R}^2 : 2x + 3y = 6\},$$

solving for y gives

$$y = -\frac{2}{3}x + 2.$$

Thus the slope is $-2/3$, the y-axis intersection is $(0, 2)$, and the x-axis intersection is $(3, 0)$.

3.1.6 The Line Through Two Points

Let

$$P = (x_1, y_1), \qquad Q = (x_2, y_2)$$

be distinct points.

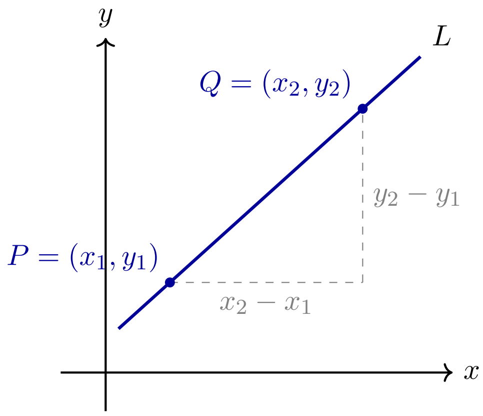
Theorem

If $x_1 \neq x_2$, the unique line through P and Q has slope

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

and can be defined by

$$L = \{(x, y) \in \mathbb{R}^2 : y - y_1 = m(x - x_1)\}.$$



The second formula follows because every point (x, y) on the line must produce the same vertical-to-horizontal change ratio from P.

If $x_1 = x_2$, the two points determine the vertical line

$$L = \{(x, y) \in \mathbb{R}^2 : x = x_1\}.$$

Example

The points $P = (1, 2)$ and $Q = (4, 8)$ give

$$m = \frac{8-2}{4-1} = 2.$$

Hence

$$y - 2 = 2(x - 1),$$

which simplifies to $y = 2x$. Therefore

$$L = \{(x, y) \in \mathbb{R}^2 : y = 2x\}.$$

3.1.7 Parallel Lines

Definition

Two distinct lines L_1 and L_2 are *parallel* when they have no common point:

$$L_1 \neq L_2 \qquad \text{and} \qquad L_1 \cap L_2 = \varnothing.$$

Consider

$$L_1 = \{(x, y) \in \mathbb{R}^2 : y = m_1 x + b_1\}, \qquad L_2 = \{(x, y) \in \mathbb{R}^2 : y = m_2 x + b_2\}.$$

A common point would have to satisfy

$$m_1 x + b_1 = m_2 x + b_2.$$

Equivalently,

$$(m_1 - m_2)x = b_2 - b_1.$$

If $m_1 \neq m_2$, this equation has exactly one solution for x, so the lines intersect. If $m_1 = m_2$, then either $b_1 = b_2$ and the lines are identical, or $b_1 \neq b_2$ and no common point exists.

Theorem

Two non-vertical lines are parallel precisely when

$$m_1 = m_2 \qquad \text{and} \qquad b_1 \neq b_2.$$

The equality of slopes is therefore a consequence of the geometric definition, not the definition itself.

Two distinct vertical lines are also parallel, because conditions $x = c_1$ and $x = c_2$ with $c_1 \neq c_2$ cannot hold at the same point.

3.1.8 Perpendicular Lines

Definition

Two intersecting lines are *perpendicular* when the angle formed at their intersection is a right angle:

$$\frac{\pi}{2}.$$

Let two non-vertical, non-horizontal lines have inclination angles α_1 and α_2. They are perpendicular exactly when their directions differ by $\pi/2$. Thus we may write

$$\alpha_2 = \alpha_1 + \frac{\pi}{2}$$

up to reversing the direction along a line.

From the previously established relation $m = \tan\alpha$,

$$m_1 = \tan\alpha_1, \qquad m_2 = \tan\alpha_2.$$

Using

$$\tan\left(\alpha + \frac{\pi}{2}\right) = -\frac{1}{\tan\alpha},$$

we obtain

$$m_2 = -\frac{1}{m_1}.$$

Theorem

For two non-vertical, non-horizontal lines,

$$L_1 \perp L_2 \iff m_1 m_2 = -1.$$

This criterion is derived from the geometric right-angle condition and the relation $m = \tan\alpha$.

A horizontal line and a vertical line form the remaining perpendicular case. The product formula is not applicable there because the vertical line has no real-valued slope.

Remark

Two lines may intersect without being perpendicular. Intersection only says that they have a common point. Perpendicularity additionally prescribes the angle at that point.

Summary

The central relations in this section are

$$\begin{aligned}
L_{m,b} &= \{(x,y) \in \mathbb{R}^2 : y = mx + b\}, \\
m &= \frac{\Delta y}{\Delta x} = \tan\alpha, \\
L_1 \parallel L_2 &\iff m_1 = m_2 \text{ and } b_1 \neq b_2, \\
L_1 \perp L_2 &\iff m_1 m_2 = -1 \quad \text{(non-vertical, non-horizontal case)}.
\end{aligned}$$

Each algebraic criterion has been obtained from a geometric definition.

3.1.9 The Parametric Equation of a Line

The equations used so far describe a line by a condition that its coordinates must satisfy. A parametric equation describes the same line differently: we start from one point and move along a fixed direction.

Let

$$P_0 = (x_0, y_0)$$

be a point of the line, and let

$$(u, v) \neq (0, 0)$$

be a direction pair. Multiplying the direction pair by a real number t changes only the distance and orientation of the displacement, not its direction. Starting at P_0 therefore gives the points

$$(x, y) = (x_0, y_0) + t(u, v), \qquad t \in \mathbb{R}.$$

Definition

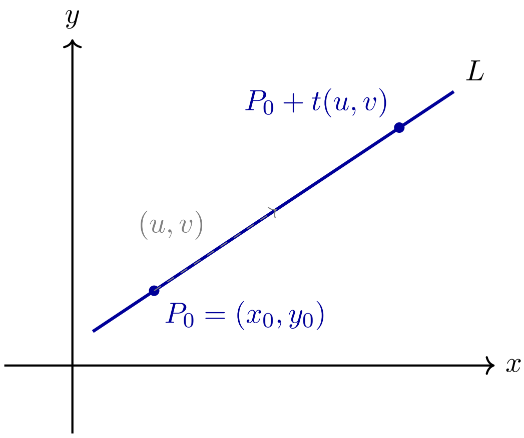
The line through $P_0 = (x_0, y_0)$ with direction pair $(u, v) \neq (0, 0)$ is parametrised by

$$L = \{(x_0 + tu,\ y_0 + tv) : t \in \mathbb{R}\}.$$

Equivalently, its coordinates satisfy

$$\begin{cases} x = x_0 + tu, \\ y = y_0 + tv, \end{cases} \qquad t \in \mathbb{R}.$$

When $t = 0$, the parametrisation gives the initial point P_0. Positive and negative values of t move from P_0 in the two opposite directions along the line. Since t ranges over all real numbers, every point of the line is obtained.



If $u \neq 0$, eliminating the parameter gives

$$t = \frac{x - x_0}{u}$$

and hence

$$y - y_0 = \frac{v}{u}(x - x_0).$$

Thus the slope is

$$m = \frac{v}{u}.$$

If $u = 0$, then $v \neq 0$ and the parametrisation becomes

$$x = x_0, \qquad y = y_0 + tv,$$

which describes a vertical line. The parametric form therefore treats non-vertical and vertical lines in one uniform way.

For two distinct points

$$P = (x_1, y_1), \qquad Q = (x_2, y_2),$$

the difference

$$Q - P = (x_2 - x_1,\, y_2 - y_1)$$

is a direction pair for their line. Consequently,

$$L = \{P + t(Q - P) : t \in \mathbb{R}\},$$

or in coordinates,

$$\begin{cases} x = x_1 + t(x_2 - x_1), \\ y = y_1 + t(y_2 - y_1), \end{cases} \qquad t \in \mathbb{R}.$$

Example

The line through $P = (1, 2)$ with direction pair $(3, 6)$ has the parametrisation

$$\begin{cases} x = 1 + 3t, \\ y = 2 + 6t, \end{cases} \qquad t \in \mathbb{R}.$$

Because $3 \neq 0$, we have

$$t = \frac{x-1}{3}.$$

Substitution into the second equation gives

$$y = 2 + 6\frac{x-1}{3} = 2x.$$

Thus the parametric and Cartesian equations describe the same line.

Remark

A parametrisation of a line is not unique. Replacing (u, v) by any non-zero multiple $\lambda(u, v)$ changes the parameter but not the set of points on the line.

3.2 Lines in Three-Dimensional Space

A point in three-dimensional Cartesian space has coordinates

$$P = (x, y, z) \in \mathbb{R}^3.$$

A line is determined by one point and one nonzero direction.

Definition

Let

$$P_0 = (x_0, y_0, z_0)$$

and let

$$\mathbf{v} = (a, b, c) \neq (0, 0, 0).$$

The line through P_0 with direction vector $\mathbf{v}$ is

$$L = \{P_0 + t\mathbf{v} : t \in \mathbb{R}\}.$$

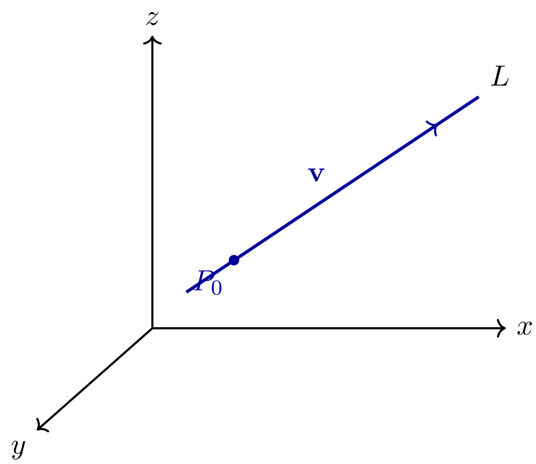
Equivalently,

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c).$$

In coordinates this becomes the system

$$\begin{cases} x = x_0 + at, \\ y = y_0 + bt, \\ z = z_0 + ct, \end{cases} \qquad t \in \mathbb{R}.$$

As t varies, the point moves along the line. Positive and negative values of t describe the two opposite directions along the same geometric line.



3.2.1 The Line Through Two Points in Space

Let

$$P = (x_1, y_1, z_1), \qquad Q = (x_2, y_2, z_2)$$

be distinct points. Their difference

$$Q - P = (x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1)$$

is a nonzero direction vector of the line through them.

Theorem

The unique line through two distinct points $P, Q \in \mathbb{R}^3$ is

$$L = \{P + t(Q - P) : t \in \mathbb{R}\}.$$

In coordinates,

$$\begin{cases} x = x_1 + t(x_2 - x_1), \\ y = y_1 + t(y_2 - y_1), \\ z = z_1 + t(z_2 - z_1). \end{cases}$$

Example

For

$$P = (1, 0, 2), \qquad Q = (3, 4, 1),$$

a direction vector is

$$Q - P = (2, 4, -1).$$

Therefore the line through P and Q is

$$(x, y, z) = (1, 0, 2) + t(2, 4, -1),$$

or

$$\begin{cases} x = 1 + 2t, \\ y = 4t, \\ z = 2 - t. \end{cases}$$

3.2.2 Symmetric Form

If a, b, and c are all nonzero, the parameter can be eliminated from

$$x = x_0 + at, \qquad y = y_0 + bt, \qquad z = z_0 + ct.$$

This gives

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

If one component of the direction vector is zero, the corresponding coordinate is constant. For example, if $c = 0$, then $z = z_0$, while the remaining two coordinates may still be written in symmetric form.

3.2.3 Parallel and Intersecting Lines in Space

Consider

$$L_1 = P_1 + t\mathbf{v}_1, \qquad L_2 = P_2 + s\mathbf{v}_2.$$

The lines are parallel when their direction vectors are proportional:

$$\mathbf{v}_2 = \lambda \mathbf{v}_1$$

for some nonzero λ.

To test whether two nonparallel lines intersect, one solves

$$P_1 + t\mathbf{v}_1 = P_2 + s\mathbf{v}_2$$

for the two parameters t and s. This is a system of three scalar equations. It may have one solution or no solution.

Definition

Two lines in $\mathbb{R}^3$ are *skew* when they are neither parallel nor intersecting.

Skew lines have no analogue in the plane. In two dimensions, two distinct nonparallel lines must intersect; in three dimensions, one line may pass above or beside another without meeting it.

Summary

A line in $\mathbb{R}^3$ is described by

$$L = \{P_0 + t\mathbf{v} : t \in \mathbb{R}\}, \qquad \mathbf{v} \neq \mathbf{0}.$$

Two distinct points determine the direction $Q - P$. Parallel lines have proportional direction vectors, while nonparallel lines may either intersect or be skew.

3.3 Planes in Three-Dimensional Space

A line in $\mathbb{R}^3$ is determined by one point and one direction. A plane requires one point and two independent directions. Equivalently, it may be described by one point and a direction perpendicular to the whole plane.

3.3.1 A Plane Through a Point with Two Direction Vectors

Let

$$P_0 = (x_0, y_0, z_0)$$

be a point and let

$$\mathbf{u} = (u_1, u_2, u_3), \qquad \mathbf{v} = (v_1, v_2, v_3)$$

be non-parallel vectors.

Definition

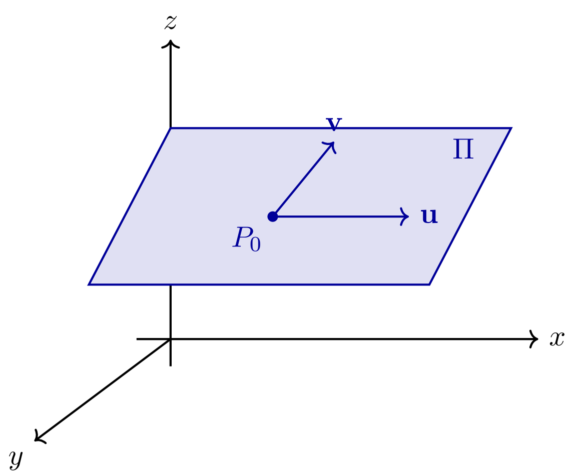
The plane through P_0 generated by $\mathbf{u}$ and $\mathbf{v}$ is

$$\Pi = \{P_0 + s\mathbf{u} + t\mathbf{v} : s, t \in \mathbb{R}\}.$$

In coordinates,

$$\begin{cases} x = x_0 + su_1 + tv_1, \\ y = y_0 + su_2 + tv_2, \\ z = z_0 + su_3 + tv_3. \end{cases}$$

The parameters s and t allow independent motion in two directions. Because $\mathbf{u}$ and $\mathbf{v}$ are not parallel, these motions fill a plane rather than a single line.



3.3.2 The Normal Vector and the Cartesian Equation

A nonzero vector

$$\mathbf{n} = (a, b, c)$$

is called a normal vector to a plane when it is perpendicular to every direction contained in that plane.

For a point $P = (x, y, z)$ on the plane, the displacement from P_0 to P is

$$P - P_0 = (x - x_0, y - y_0, z - z_0).$$

This displacement lies in the plane and is therefore perpendicular to $\mathbf{n}$. Hence

$$\mathbf{n} \cdot (P - P_0) = 0.$$

Theorem

A plane through $P_0 = (x_0, y_0, z_0)$ with normal vector $\mathbf{n} = (a, b, c) \neq (0, 0, 0)$ has equation

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

After collecting constants, this becomes

$$ax + by + cz = d$$

for some $d \in \mathbb{R}$.

Thus a first-degree equation in three variables describes a plane, just as a first-degree equation in two variables describes a line.

Example

The plane through $P_0 = (1, -1, 2)$ with normal vector $\mathbf{n} = (2, 1, -3)$ satisfies

$$2(x - 1) + (y + 1) - 3(z - 2) = 0.$$

Therefore

$$2x + y - 3z = -5.$$

3.3.3 A Plane Through Three Points

Let P, Q, and R be three points that do not lie on one line. Then

$$\mathbf{u} = Q - P, \qquad \mathbf{v} = R - P$$

are non-parallel direction vectors in the plane.

Theorem

Three non-collinear points determine exactly one plane. A parametric equation is

$$\Pi = \{P + s(Q - P) + t(R - P) : s, t \in \mathbb{R}\}.$$

A normal vector may be obtained from the cross product

$$\mathbf{n} = (Q - P) \times (R - P).$$

Example

For

$$P = (1, 0, 0), \qquad Q = (0, 1, 0), \qquad R = (0, 0, 1),$$

we have

$$Q - P = (-1, 1, 0), \qquad R - P = (-1, 0, 1).$$

Their cross product is

$$(Q - P) \times (R - P) = (1, 1, 1).$$

Using the point P, the plane equation is

$$(x-1)+y+z=0,$$

so

$$x+y+z=1.$$

3.3.4 Relative Positions of Planes

Consider

$$\Pi_1 : a_1x+b_1y+c_1z=d_1, \qquad \Pi_2 : a_2x+b_2y+c_2z=d_2,$$

with normal vectors

$$\mathbf{n}_1=(a_1,b_1,c_1), \qquad \mathbf{n}_2=(a_2,b_2,c_2).$$

Theorem

Two distinct planes are parallel precisely when their normal vectors are parallel. If the normal vectors are not parallel, the planes intersect in a line.

The planes are perpendicular when their normal vectors are perpendicular:

$$\mathbf{n}_1\cdot\mathbf{n}_2=0.$$

3.3.5 A Line as the Intersection of Two Planes

In $\mathbb{R}^3$, two non-parallel planes usually meet in a line. Thus a line may also be described by a system

$$\begin{cases} a_1x+b_1y+c_1z=d_1, \\ a_2x+b_2y+c_2z=d_2. \end{cases}$$

A direction vector for the intersection line is perpendicular to both normal vectors, so one may take

$$\mathbf{v}=\mathbf{n}_1\times\mathbf{n}_2.$$

Summary

A plane in $\mathbb{R}^3$ may be described in two fundamental ways:

$$P=P_0+s\mathbf{u}+t\mathbf{v}$$

with two non-parallel direction vectors, or

$$ax+by+cz=d$$

with a nonzero normal vector (a,b,c). Three non-collinear points determine one plane, and two non-parallel planes intersect in a line.

Chapter 4

Second-Degree Equations, Conic Sections, and Quadric Surfaces

4.1 Second-Degree Equations and Conic Sections

Lines arise from equations in which the coordinates occur only to the first power. The next natural class consists of equations in which terms of total degree two are allowed.

Definition

A *second-degree equation* in the Cartesian plane has the form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

where $A, B, C, D, E, F \in \mathbb{R}$ and at least one of A, B, or C is nonzero. Its solution set is the collection of all points $(x, y) \in \mathbb{R}^2$ whose coordinates satisfy the equation.

The terms

$$Ax^2 + Bxy + Cy^2$$

form the quadratic part. The remaining terms affect the position and size of the curve, while the quadratic part largely determines its geometric type.

4.1.1 A First Classification

For the equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

define

$$\Delta = B^2 - 4AC.$$

Theorem

For a non-degenerate real conic, the sign of Δ gives the following classification:

$$\begin{array}{r|c} \Delta < 0 & \text{ellipse type, including circles,} \\ \Delta = 0 & \text{parabola type,} \\ \Delta > 0 & \text{hyperbola type.} \end{array}$$

This classification is preserved by rotations and translations of the coordinate system. A rotation can remove the mixed term Bxy, while a translation can move the centre or vertex to a simpler position.

Remark

The sign of Δ identifies the conic type only after degenerate cases are separated. A second-degree equation may also describe a single point, no real points, one line, or two lines. These possibilities are discussed below.

4.1.2 Circle

A circle with centre (h, k) and radius $r > 0$ is described by

$$(x - h)^2 + (y - k)^2 = r^2.$$

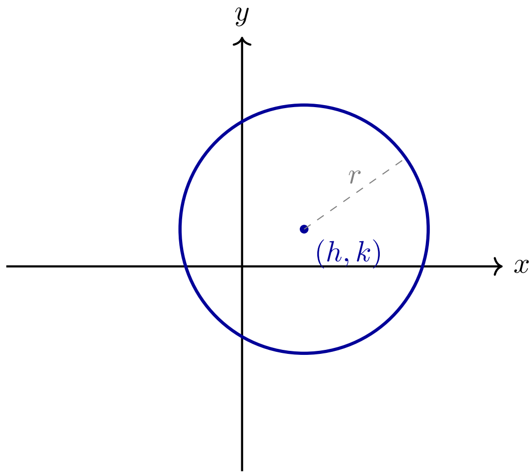
After expansion,

$$x^2 + y^2 - 2hx - 2ky + h^2 + k^2 - r^2 = 0.$$

Here $A = C = 1$ and $B = 0$, so

$$\Delta = 0^2 - 4 \cdot 1 \cdot 1 = -4 < 0.$$

A circle is therefore a special ellipse-type conic in which the coefficients of x^2 and y^2 are equal and there is no mixed term after a suitable rotation.



4.1.3 Ellipse

The standard ellipse centred at the origin has equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad a > 0, \quad b > 0.$$

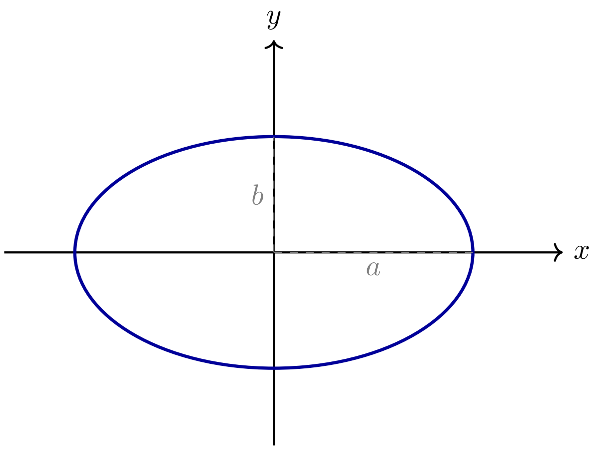
Equivalently,

$$b^2x^2 + a^2y^2 - a^2b^2 = 0.$$

Thus $A = b^2$, $B = 0$, and $C = a^2$, so

$$\Delta = -4a^2b^2 < 0.$$

The parameters a and b are the semi-axis lengths. When $a = b$, the ellipse becomes a circle.



4.1.4 Parabola

A standard vertical parabola has equation

$$y = ax^2, \qquad a \neq 0,$$

or equivalently

$$ax^2 - y = 0.$$

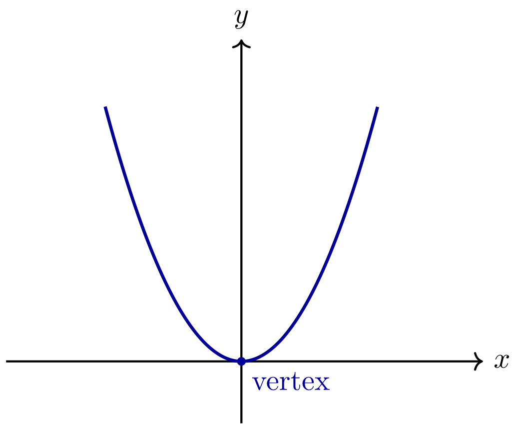
Here $A = a$, $B = 0$, and $C = 0$, hence

$$\Delta = 0.$$

The sign of a determines whether the parabola opens upward or downward. A translated form is

$$y - k = a(x - h)^2,$$

whose vertex is (h, k).



4.1.5 Hyperbola

A standard hyperbola centred at the origin has equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \qquad a > 0, \quad b > 0.$$

Equivalently,

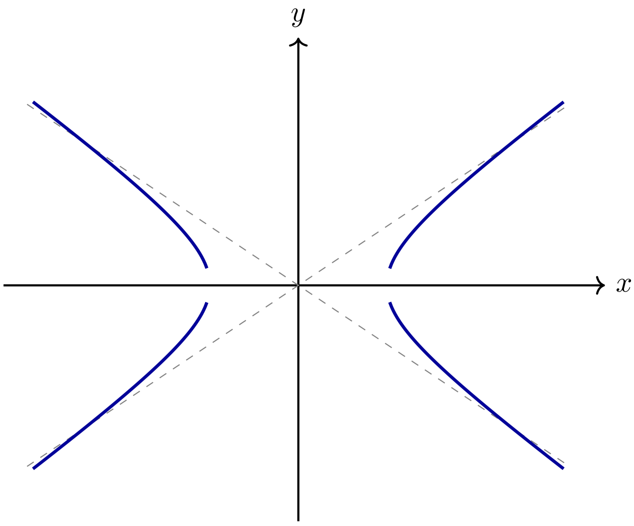
$$b^2x^2 - a^2y^2 - a^2b^2 = 0.$$

Here $A = b^2$, $B = 0$, and $C = -a^2$, so

$$\Delta = 4a^2b^2 > 0.$$

Its asymptotes are

$$y = \frac{b}{a}x, \qquad y = -\frac{b}{a}x.$$



4.1.6 The Mixed Term and Rotation

The term Bxy indicates that the natural axes of the conic may be rotated relative to the chosen Cartesian axes. For example,

$$x^2 + 2xy + y^2 = 1$$

can be written as

$$(x + y)^2 = 1,$$

so it is not a non-degenerate ellipse: it describes the two parallel lines

$$x + y = 1, \qquad x + y = -1.$$

For a genuine rotated conic, one introduces new coordinates (u, v) obtained by a rotation through an angle θ:

$$x = u\cos\theta - v\sin\theta, \qquad y = u\sin\theta + v\cos\theta.$$

Choosing θ appropriately removes the uv term. When $A \neq C$, one may choose θ so that

$$\tan(2\theta) = \frac{B}{A - C}.$$

The resulting equation can then be classified in the new coordinates using the standard forms above.

4.1.7 Degenerate Second-Degree Equations

Not every second-degree equation describes a non-degenerate conic. Factorisation or completing squares may reveal a simpler set.

Example

The equation

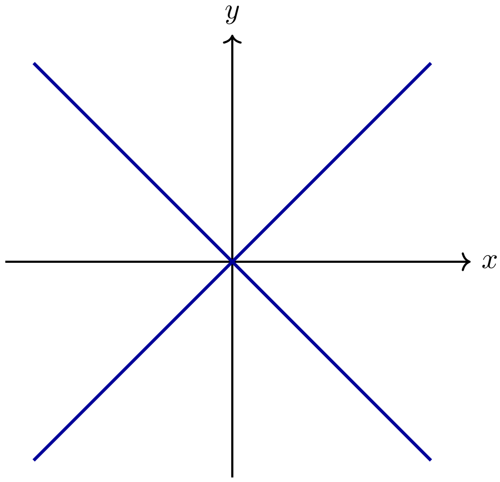
$$x^2 - y^2 = 0$$

factors as

$$(x - y)(x + y) = 0.$$

Its solution set is the union of two intersecting lines:

$$y = x \qquad \text{or} \qquad y = -x.$$



Other common degenerate cases include

$$x^2 + y^2 = 0,$$

which gives only the point $(0, 0)$,

$$x^2 + y^2 + 1 = 0,$$

which has no real solutions, and

$$x^2 = 0,$$

which describes the single line $x = 0$ counted algebraically twice.

Summary

Second-degree equations have the general form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

For non-degenerate real conics, the discriminant

$$\Delta = B^2 - 4AC$$

separates the three principal types:

$\Delta < 0$ gives ellipse type, $\Delta = 0$ gives parabola type, $\Delta > 0$ gives hyperbola type.

Translations and rotations reduce many equations to standard forms, while factorisation and completing squares expose degenerate cases.

4.2 Second-Degree Surfaces in Three-Dimensional Space

The three-dimensional analogue of a conic section is a *quadric surface*. Instead of an equation in two coordinates, we now allow a second-degree equation in three coordinates.

Definition

A second-degree equation in $\mathbb{R}^3$ has the form

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0,$$

where at least one quadratic coefficient is nonzero. Its solution set is called a quadric surface when it forms a non-degenerate surface.

Translations move the centre or vertex of the surface. Rotations of the coordinate system remove mixed terms such as xy, xz, and yz. After these changes of coordinates, the principal quadric surfaces have simple standard forms.

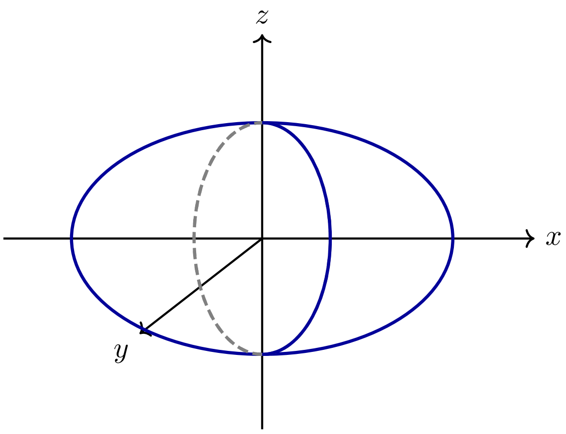
4.2.1 Ellipsoid

An ellipsoid centred at the origin has equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \qquad a, b, c > 0.$$

Its intersections with the coordinate planes are ellipses. When $a = b = c = r$, the ellipsoid becomes the sphere

$$x^2 + y^2 + z^2 = r^2.$$

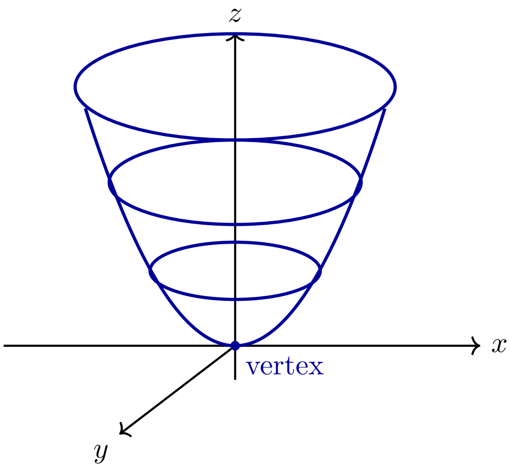


4.2.2 Elliptic Paraboloid

An elliptic paraboloid has standard equation

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}, \qquad a, b > 0.$$

Horizontal sections $z = k > 0$ are ellipses, while vertical sections through the z-axis are parabolas. The surface has one vertex and opens in one direction.

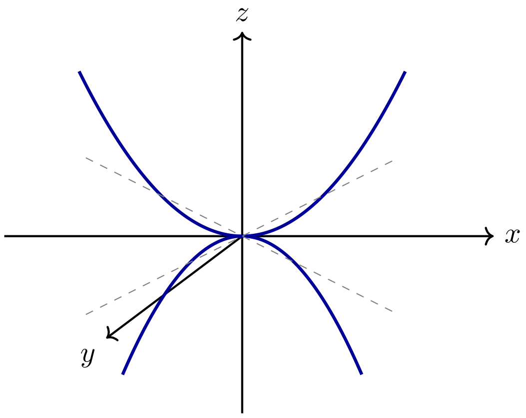


4.2.3 Hyperbolic Paraboloid

A hyperbolic paraboloid has standard equation

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}.$$

Sections parallel to the xz-plane and yz-plane are parabolas opening in opposite directions. Horizontal sections are hyperbolas, except at $z = 0$, where the section consists of two intersecting lines. The surface is commonly called a saddle.

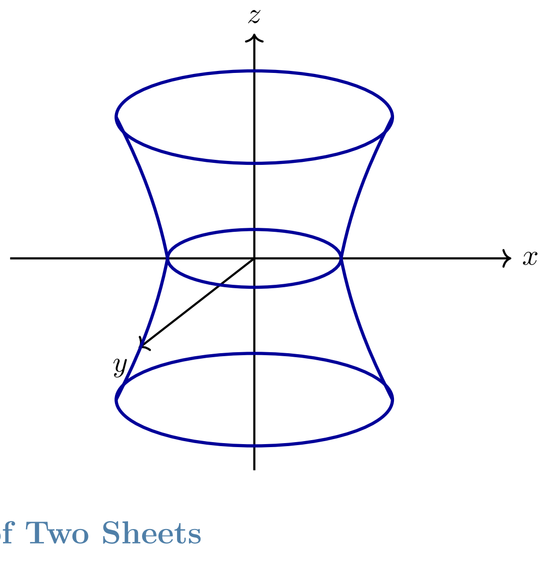


4.2.4 Hyperboloid of One Sheet

A hyperboloid of one sheet has standard equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1.$$

Horizontal sections are ellipses. Vertical sections through the z-axis are hyperbolas. The surface is connected and has a narrow central waist.

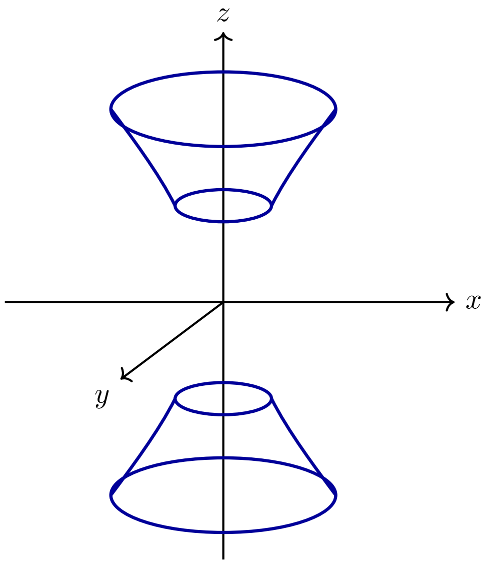


4.2.5 Hyperboloid of Two Sheets

A hyperboloid of two sheets has standard equation

$$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

The condition forces $|z| \geq c$, so the surface has two disconnected parts. Horizontal sections are ellipses, while suitable vertical sections are hyperbolas.

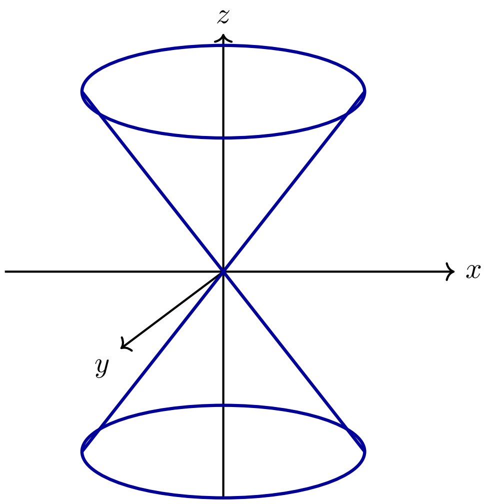


4.2.6 Elliptic Cone

An elliptic cone has equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0.$$

It consists of two nappes meeting at the origin. Horizontal sections away from the origin are ellipses. Vertical sections through the axis are pairs of intersecting lines.



4.2.7 Cylinders as Second-Degree Surfaces

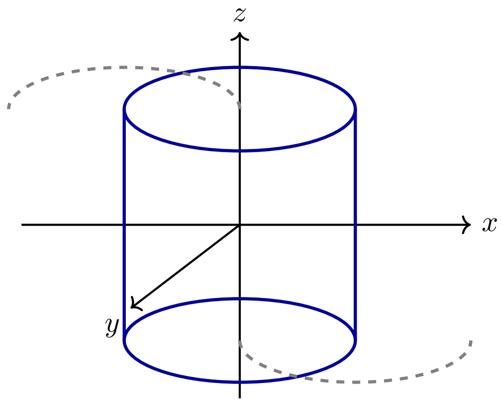
If one coordinate does not occur in the equation, the corresponding planar curve is extended parallel to that coordinate axis. For example,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

describes an elliptic cylinder parallel to the z-axis, while

$$y = x^2$$

describes a parabolic cylinder parallel to the z-axis.



4.2.8 Degenerate Quadrics

As in the planar case, a second-degree equation in three variables may be degenerate. It may describe a point, a line, a plane, two planes, or no real points. For example,

$$x^2 + y^2 + z^2 = 0$$

describes only the origin, while

$$x^2 - y^2 = 0$$

factors as

$$(x - y)(x + y) = 0$$

and therefore describes the union of the two planes

$$x - y = 0 \qquad \text{and} \qquad x + y = 0.$$

Summary

Second-degree equations in three variables produce quadric surfaces. After translations and rotations, the main non-degenerate types are ellipsoids, elliptic and hyperbolic paraboloids, hyperboloids of one or two sheets, and elliptic cones. Equations missing one coordinate give cylinders, while factorisation may reveal degenerate unions of planes or smaller sets.

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